



Discrete Structures

CSC242

Covenant Centre for Open
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Module One

Descriptive Statistics

Unit 1 — Descriptive Statistics

INTRODUCTION

Logic will get you from A to B. Imagination will take you everywhere.” – Albert Einstein

Logic is a formal study of mathematics; it is the study of mathematical reasoning and proof itself.

The rules of logic give precise meaning to mathematical statements. These rules are used to distinguish between valid and invalid mathematical arguments. Apart from its importance in understanding mathematical reasoning, logic has numerous applications in Computer Science, varying from the design of digital circuits to the construction of computer programs and the verification of the correctness of programs.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Identify and construct well-formed propositional formulas using logical connectives.
- Apply truth tables to determine the validity, satisfiability, and equivalence of propositions.
- Translate real-world statements into propositional logic expressions.

1.1.1 Proposition and Propositional Logic

Perhaps the most distinguishing characteristic of mathematics is its reliance on logic. Explicit training in mathematical logic is essential to a mature understanding of mathematics. Familiarity with the concepts of logic is also a prerequisite to studying several central areas of computer science, including databases, compilers, and complexity theory

The simplest variant of formal logic is propositional logic. Its basic object is a simple, declarative sentence, called a proposition. Propositional logic is concerned with combining sentences, such as "The world is round" and "Columbus was right" to form "If the world is round, then Columbus was right."

A proposition is something that is either true or false; it is not both. "The cover of this book is pink" is a proposition. "Napoleon spent at least one day of his life in Paris" and "Either the butler did it with a bottle or the colonel did it with a lead pipe" are also propositions. On the other hand, "Justice," "The Queen's birthday," "Whoever is the stronger," and "Why is the world almost round?" are neither true nor false and, therefore, are not propositions.

Propositional Logic is the logical system built around propositions. From such propositions, one can build logical arguments and implications

Introduction to Propositional Logic

Differential equations can be classified based on: The simplest variant of formal logic is propositional logic. Its basic object is a simple, declarative sentence, called a proposition. Propositional logic is concerned with combining sentences, such as "The world is round" and "Columbus was right" to form "If the world is round, then Columbus was right."

A proposition is something that is either true or false; it is not both. "The cover of this book is pink" is a proposition. "Napoleon spent at least one day of his life in Paris" and "Either the butler did it with a bottle or the colonel did it with a lead pipe" are also propositions. On the other hand, "Justice," "The Queen's birthday," "Whoever is the stronger," and "Why is the world almost round?" are neither true nor false and, therefore, are not propositions.

In formal notation, the letters p , q , r , and s (plus those letters subscripted with natural numbers, such as p_1 , q_2 , and r_{127}) are used to stand for, or to denote, propositions. Such a variable is called a proposition letter. We consider proposition letters to be essentially the same as boolean (logical) variables in a programming language.

T and F are propositional constants-that is, propositions with fixed truth values of TRUE and FALSE, respectively.

The propositions are combined using **Logical Connectives** or **Logical Operators**

Table 1: Propositional Logic

Connective Symbols	Words	Terms	Example
$\wedge$	AND	Conjunction	$A \wedge B$
$\vee$	OR	Disjunction	$A \vee B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\Leftrightarrow$	If and only If	Biconditional	$A \Leftrightarrow B$
$\neg$	Not	Negation	$\neg A$

Truth Table of Propositional Logic

Since we need to know the truth value of a proposition in all possible scenarios, we consider all the possible combinations of the propositions which are joined together by Logical Connectives to form the given compound proposition. This compilation of all possible scenarios in a tabular format is called a **truth table**. Most Common Logical Connectives-

1. Negation

If p is a proposition, then the negation of p is denoted by $\neg p$, which, when translated to simple English, means "It is not the case that **p**" or simply "not **p**". The truth value of **-p** is the opposite of the truth value of **p**. The truth table of **-p** is:

p	$\neg p$
T	F
F	T

Example, Negation of "It is raining today", is "It is not the case that is raining today" or simply "It is not raining today".

2. Conjunction

For any two propositions p and q , their conjunction is denoted by $p \wedge q$, which means " p and q ". The conjunction $p \wedge q$ is True when both p and q are True, otherwise False. The truth table of $p \wedge q$ is:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example, Conjunction of the propositions p - "Today is Friday" and q - "It is raining today", $p \wedge q$ is "Today is Friday, and it is raining today". This proposition is true only on rainy Fridays and is false on any other rainy day or on Fridays when it does not rain.

3. Disjunction

For any two propositions p and q , their disjunction is denoted by $p \vee q$, which means " p or q ". The disjunction $p \vee q$ is True when either p or q is True, otherwise False. The truth table of $p \vee q$ is:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example, Disjunction of the propositions p - "Today is Friday" and q - "It is raining today", $p \vee q$ is "Today is Friday or it is raining today". This proposition is true on any day that is a Friday or a rainy day (including rainy Fridays) and is false on any day other than Friday when it also does not rain.

4. Exclusive Or

For any two propositions p and q , their exclusive or is denoted by $p \oplus q$, which means "either p or q but not both". The exclusive or $p \oplus q$

is True when either p or q is True, and False when both are true or both are false. The truth table of $p \oplus q$ is:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Example, Exclusive or of the propositions p - "Today is Friday" and q - "It is raining today", $p \oplus q$ is "Either today is Friday or it is raining today, but not both". This proposition is true on any day that is a Friday or a rainy day (not including rainy Fridays) and is false on any day other than Friday when it does not rain or on rainy Fridays.

5. Implication

For any two propositions p and q , the statement "if p then q " is called an implication, and it is denoted by $p \rightarrow q$. In the implication $p \rightarrow q$, p is called the **hypothesis** or **antecedent** or **premise** and q is called the **conclusion** or **consequence**. The implication $p \rightarrow q$ is also called a **conditional statement**. The implication is false when p is true and q is false; otherwise, it is true. The truth table of $p \rightarrow q$ is:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

One might wonder that why is $p \rightarrow q$ true when p is false. This is because the implication guarantees that when p and q are true then the implication is true. But the implication does not guarantee anything when the premise p is false. There is no way of knowing whether or not the implication is false since p did not happen. This situation is similar to the "Innocent until proven Guilty" stance, which means that the implication $p \rightarrow q$ is considered true until

proven false. Since we cannot call the implication $p \rightarrow q$ false when p is false, our only alternative is to call it true.

This follows from the **Explosion Principle**, which says: "A False statement implies anything" Conditional statements play a very important role in mathematical reasoning; thus, a variety of terminology is used to express $p \rightarrow q$, some of which are listed below.

"If p , then q " " p is sufficient for q " " q when p " " a necessary condition for p is q " " p only if q " " q unless $\neg p$ " " q follows from p "

Example, "If it is Friday, then it is raining today" is a proposition which is of the form $p \rightarrow q$. The above proposition is true if it is not Friday (premise is false) or if it is Friday and it is raining, and it is false when it is Friday but it is not raining.

6. Biconditional or Double Implication

For any two propositions p and q , the statement " p if and only if (iff) q " is called a biconditional and it is denoted by $p \leftrightarrow q$. The statement $p \leftrightarrow q$ is also called a **bi-implication**. $p \leftrightarrow q$ has the same truth value as $(p \rightarrow q) \wedge (q \rightarrow p)$. The implication is true when p and q have same truth values, and is false otherwise. The truth table of $p \leftrightarrow q$ is:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Some other common ways of expressing $p \leftrightarrow q$ are:

" p is necessary and sufficient for q " " $\text{if } p \text{ then } q$, and conversely" " p if q "

Example, "It is raining today if and only if it is Friday today." is a proposition which is of the form $p \leftrightarrow q$. The above proposition is true if it is not Friday and it is not raining or if it is Friday and it is raining, and it is false when it is not Friday or it is not raining

In-Text Questions (ITQs)

Consider the following statements:

P: Good mobile phones are not cheap.

Q: Cheap mobile phones are not good.

L: P implies Q

M: Q implies P

N: P is equivalent to Q

Which one of the following about L, M, and N is CORRECT?

- a. Only L is TRUE.
- b. Only M is TRUE.
- c. Only N is TRUE.
- d. L, M and N are TRUE.

In-Text Answers (ITAs)

Let a and b be two propositions

a: Good Mobile phones.

b: Cheap Mobile Phones.

P and Q can be written in logic as

P: $a \rightarrow \sim b$

Q: $b \rightarrow \sim a$.

Truth Table

a	b	$\sim a$	$\sim b$	P	Q
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

it clearly shows P and Q are equivalent.

so option D is Correct.

Propositional Connectives

Propositional logic is concerned with certain ways in which simple sentences can be combined into more complex sentences. Several standard operations are used on propositions to form other propositions. Such an operation is called a propositional connective. The common propositional connectives are shown in Table 2.

Connective	Sample Use	Common Translation
$\neg$	$\neg p$	"not p "
$\wedge$	$p \wedge q$	" p and q "
$\vee$	$p \vee q$	" p or q (or both)"
$\rightarrow$	$p \rightarrow q$	"if p , then q ," or " p implies q "
$\leftrightarrow$	$p \leftrightarrow q$	" p if and only if q ," or " p is equivalent to q "

Example. Let p denote "Henry eats halibut" and q denote "Catherine eats kippers."

- The proposition $\neg p$ is read "Henry does not eat halibut."
- The proposition $p \wedge q$ is read "Henry eats halibut, and Catherine eats kippers."
- The proposition $p \rightarrow q$ is read "If Henry eats halibut, then Catherine eats kippers."
- The proposition $p \leftrightarrow q$ is read "Henry eats halibut if and only if Catherine eats kippers."
- The proposition $(\neg p) \vee (\neg q)$ is read "Henry does not eat halibut, or Catherine does not eat kippers."
- The proposition $p \leftrightarrow (\neg q)$ is read "Henry eats halibut if and only if Catherine does not eat kippers."

Example. Let p denote "Henry eats halibut," q denote "Catherine eats kippers," and r denote "I'll eat my hat."

- Write a proposition that reads "If Henry eats halibut but Catherine does not eat kippers, then I'll eat my hat."
- Write a proposition that reads "Either Henry eats halibut or Catherine eats kippers, but not both."

Solution.

- $(p \wedge \neg q) \rightarrow r$. Since *and* and *but* usually both get translated as $\wedge$, the difference between the two English words is usually an issue not of what is the case but, rather, of what we would have expected to be the case.

(b) $(p \vee q) \wedge \neg (p \wedge q)$. This proposition is "logically equivalent to" the proposition in Example 1 (f), meaning that $p \leftrightarrow (\neg q)$ is an equally good answer. We shall discuss logical equivalence in the next section.

Conjunction of p and q , denoted by $p \wedge q$, is the proposition 'p and q'. The conjunction $p \wedge q$ is True, when both p and q is True. **Disjunction** of p and q , denoted by $p \vee q$, is the proposition 'p or q'. The disjunction $p \vee q$ is False when both p and q is False.

Logical Implication - It is a type of relationship between two statements or sentences. Denoted by ' $p \rightarrow q$ '. The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. i.e. $p \rightarrow q = \neg p \vee q$

Bi-Condition A bi-conditional statement is a compound statement formed by combining two conditionals under "and." Bi-conditionals are true when both statements have the same truth value.

Example: The atomic statements: "It is raining" and "I need an umbrella" can be represented by the letters P and Q , respectively. Draw a truth table for the atomic statements.

Solution

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	F	T	T	F
F	F	T	T	F	F	T	T

Example: Make a truth table for the statement $\neg P \vee Q$.

Solution

In solving such exercises, you will have to be careful as to knowing the exact position of the $\neg$. Note that this statement is not $\neg(P \vee Q)$, the negation belongs only to P (i.e. $\neg P$). Here is the truth table:

P	Q	$\neg P$	$\neg P \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

In-Text Questions (ITQs)

Analyze the statement, "if you get more doubles than any other player you will lose, or that if you lose you must have bought the most properties," using truth tables

In-Text Answers (ITAs)

Let's start by breaking down the molecular statement into atomic statements. Let P be the statement "you get more doubles than any other player,"; Q be the statement "you will lose," and R be the statement "you must have bought the most properties." Now let's construct a truth table to represent the statement as this symbol $(P \rightarrow Q) \vee (Q \rightarrow R)$.

The truth table needs to contain 8 rows in order to account for every possible combination of truth and falsity among the three statements. Here is the full truth table:

P	Q	R	$(P \rightarrow Q)$	$(Q \rightarrow R)$	$(P \rightarrow Q) \vee (Q \rightarrow R)$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

This is a true statement about monopoly, such that it is regardless of how many properties you own, how many doubles you roll, or whether you win or lose, the outcome is true for all 8 possible combinations.

1.1.2 Conclusion

Understanding propositional logic is essential in computer science and mathematics. It enhances our ability to form valid arguments, build precise logical expressions, and analyse truth values systematically. By mastering the use of logical connectives and truth tables, learners are better prepared for advanced topics in programming, data science, and formal reasoning.

Summary

This unit introduced Propositional Logic as a foundation of mathematical reasoning. It discussed propositions, logical connectives, and truth tables. Learners were guided through key logical operations; negation, conjunction, disjunction, exclusive or, implication, and biconditional, with examples and truth table constructions. The unit emphasized logical equivalence and translation of real-world statements into logical expressions, highlighting the relevance of logic in computing and mathematical proofs

Self-Assessment Questions

1. What is the difference between a proposition and a non-propositional statement? Provide two examples of each.
2. Construct a truth table for the compound statement: $\neg(p \wedge q) \rightarrow (r \vee p)$.
3. How does the implication $p \rightarrow q$ differ from the biconditional $p \leftrightarrow q$ in terms of truth values?

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. Tutor Marked Assignment: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. Given the propositions:

p: "The network is secure"

q: "The firewall is active"

r: "The system is not under attack"

Write and interpret the logical expression:

$$(p \wedge q) \rightarrow r(p \wedge q) \rightarrow r(p \wedge q) \rightarrow r.$$

Then, construct its truth table. Explain the differences between first-order and second-order differential equations with examples.

2. Translate the following real-world statement into propositional logic and construct its truth table:

"If it is sunny or windy, then the drone will not fly."

Use:

p: "It is sunny"

q: "It is windy"

r: "The drone will fly"

References/Further Readings

Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.

Epp, S. S. (2011). Discrete Mathematics with Applications (4th ed.). Cengage Learning.

Gries, D., & Schneider, F. B. (1993). A Logical Approach to Discrete Math. Springer

Module Two

Predicate Logic

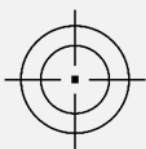
Unit 1 — Predicate Logic

INTRODUCTION

In this unit, we will introduce a more powerful type of logic called predicate logic. We will see how predicate logic can be used to express the meaning of a wide range of statements in mathematics and computer science in ways that permit us to reason and explore relationships between objects. To understand predicate logic, we first need to introduce the concept of a predicate. Afterward, we will introduce the notion of quantifiers, which enable us to reason with statements that assert that a certain property holds for all objects of a certain type and with statements that assert the existence of an object with a particular property

Statements involving variables, such as " $x > 3$," " $x = y + 3$," " $x + y = z$," and "computer x is under attack by an intruder," and "computer x is functioning properly," are often found in mathematical assertions, in computer programs, and in system specifications.

These statements are neither true nor false when the values of the variables are not specified. In this section, we will discuss the ways that propositions can be produced from such statements.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Distinguish between propositions and predicates in formal logic.
- Apply universal ($\forall$) and existential ($\exists$) quantifiers in logical expressions.
- Translate natural language statements into predicate logic formulas.
- Construct and evaluate truth values of predicate logic expressions involving quantifiers.

2.1.1 Definition

What is logic?

It is a branch of mathematics that deals with the verification of truth/falsity of mathematical statements/assertions

It is the study of valid arguments, inference (induction + deduction), proofs, proof techniques, proving theorems, paradoxes, and fallacies

"Logic will get you from A to B. Imagination will take you everywhere." A. Einstein

What is a predicate?

A predicate is neither true nor false.

A predicate becomes a proposition when the variables are substituted with specific values.

The domain of a predicate variable is the set of all values that may be substituted for the variable.

A proposition is a statement that is either true or false.

A propositional function or predicate is a sentence that contains one or more variables.

A compound proposition can be created from other propositions using logical connectives

The truth of a compound proposition is defined by the truth values of elementary propositions and the meaning of connectives.

The truth table for a compound proposition: a table with entries (rows) for all possible combinations of truth values of elementary propositions.

2.1.2 Propositional Logic

Propositional logic: a formal language for making logical inference

Example 1:

All the following declarative sentences are propositions.

1. Washington, D.C., is the capital of the United States of America.
2. Toronto is the capital of Canada.

3. $1 + 1 = 2$.

4. $2 + 2 = 3$.

Propositions 1 and 3 are true, whereas 2 and 4 are false.

Example

Symbol	Predicate	Domain	Propositions
$p(x)$	$x > 5$	$x \in \mathbb{R}$	$p(6), p(-3.6), p(0), \dots$
$p(x, y)$	$x + y$ is odd	$x \in \mathbb{Z}, y \in \mathbb{Z}$	$p(4, 5), p(-4, -4), \dots$
$p(x, y)$	$x^2 + y^2 = 4$	$x \in \mathbb{R}, y \in \mathbb{R}$	$p(-1.7, 8.9), p(-\sqrt{3}, -1), \dots$

Propositional logic is concerned with certain ways in which simple sentences can be combined into more complex sentences. Several standard operations are used on propositions to form other propositions. Such an operation is called a propositional connective. The common propositional connectives are shown in Table below

Table: Propositional Connectives

Connective	Sample Use	Common Translation
$\neg$	$\neg p$	"not p "
$\wedge$	$p \wedge q$	" p and q "
$\vee$	$p \vee q$	" p or q (or both)"
$\rightarrow$	$p \rightarrow q$	"if p , then q ," or " p implies q "
$\leftrightarrow$	$p \leftrightarrow q$	" p if and only if q ," or " p is equivalent to q "

Usefulness of Predicate Logic for Natural Language Semantics

- While in propositional logic, we can only talk about sentences as a whole, predicate logic allows us to decompose simple sentences into smaller parts: predicates and individuals.

(1)

a. John is tall.

b. $T(j)$

- Predicate logic provides a tool to handle expressions of generalization: i.e., quantificational expressions.

(2)

a. Every cat is sleeping.

b. Some girl likes David.

c. No one is happy.

- Predicate logic allows us to talk about variables (pronouns). The value for the pronoun is some individual in the domain of universe that is contextually determined.
- (3)
- a. It is sleeping.
 - b. She likes David.
 - c. He is happy.
- Sentences with quantificational expressions can be divided into two interpretive components.
- (4) Every cat is sleeping.
- a. (every cat)(it is sleeping)
 - b. for all x , x is a cat, x is sleeping
 - c. = true iff 'it is sleeping' is true for all possible values for 'it' in the domain

A simple sentence containing a variable/pronoun (a place holder): It can be evaluated as true or false with respect to some individual contextually taken as a value for the pronoun.

Truth Table

Since we need to know the truth value of a proposition in all possible scenarios, we consider all the possible combinations of the propositions which are joined together by Logical Connectives to form the given compound proposition. This compilation of all possible scenarios in a tabular format is called a truth table.

Most Common Logical Connectives-

1. Negation

If p is a proposition, then the negation of p is denoted by $\neg p$, which, when translated to simple English, means "It is not the case that p " or simply "not p ". The truth value of $\neg p$ is the opposite of the truth value of p . The truth table of $\neg p$ is:

p	$\neg p$
T	F
F	T

Example, Negation of "It is raining today", is "It is not the case that is raining today" or simply "It is not raining today".

2. Conjunction

For any two propositions p and q , their conjunction is denoted by $p \wedge q$, which means "p and q". The conjunction $p \wedge q$ is True when both p and q are True, otherwise False. The truth table of $p \wedge q$ is:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

3. Disjunction

For any two propositions p and q , their disjunction is denoted by $p \vee q$, which means "p or q". The disjunction $p \vee q$ is True when either p or q is True, otherwise False. The truth table of $p \vee q$ is:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

4. Exclusive Or

For any two propositions p and q , their exclusive or is denoted by $p \oplus q$, which means "either p or q but not both". The exclusive or $p \oplus q$ is True when either p or q is True, and False when both are true or both are false. The truth table of $p \oplus q$ is:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

5. Implication

For any two propositions p and q , the statement "if p then q " is called an implication and it is denoted by $p \rightarrow q$. In the implication $p \rightarrow q$, p is called the hypothesis or antecedent or premise and q is called the conclusion or consequence. The implication $p \rightarrow q$ is also called a conditional statement. The implication is false when p is true, and q is false; otherwise, it is true. The truth table of $p \rightarrow q$ is:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

6. Biconditional or Double Implication

For any two propositions p and q , the statement " p if and only if (iff) q " is called a biconditional and it is denoted by $p \leftrightarrow q$. The statement $p \leftrightarrow q$ is also called a bi-implication. $p \leftrightarrow q$ has the same truth value as $(p \rightarrow q) \wedge (q \rightarrow p)$. The implication is true when p and q have same truth values, and is false otherwise. The truth table of $p \leftrightarrow q$ is:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

The quantificational expression:

It instructs us to limit the domain of individuals being considered to a relevant set (e.g., a set of cats), and tells us how many different values of the pronoun we have to consider from that domain to establish truth for the sentence.

(5) Some girls like David.

a. (some girl)(she likes David)

b. for some x , x is a girl, x likes David

c. = true iff 'she likes David' is true for at least one possible value for 'she' "Logic will get you from A to B. Imagination will take you everywhere." A. Einstein

What is a truth set?

A truth set of a predicate is the set of all values of the predicate that make the predicate true.

If $p(x)$ is a predicate and x has domain D , then the truth set of $p(x)$ is the set of all elements of D that make $p(x)$ true when the values are substituted for x . That is

Truth set of $p(x) = \{x \in D \mid p(x)\}$

Example

Symbol	Predicate	Domain	Truth set
$p(x)$	$x > 5$	$x \in \mathbb{R}$	$\{p(6), p(13.6), p(5.001), \dots\}$
$p(x, y)$	$x + y$ is odd	$x \in \mathbb{Z}, y \in \mathbb{Z}$	$\{p(4, 5), p(-4, -3), \dots\}$
$p(x, y)$	$x^2 + y^2 = 4$	$x \in \mathbb{R}, y \in \mathbb{R}$	$\{p(-2, 2), p(-\sqrt{3}, -1), \dots\}$

A proposition is a statement that is either true or false.

Examples: –

Pitt is located in the Oakland section of Pittsburgh.

$5 + 2 = 8$.

It is raining today

2 is a prime number

If (you do not drive over 65 mph), then (you will not get a speeding ticket).

The following are not propositions:

– How are you?

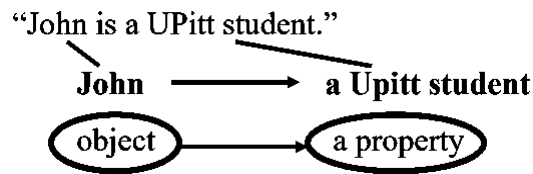
– $x + 5 = 3$

Limitations of propositional logic

In Propositional logic, the world is described in terms of elementary propositions and their logical combinations

Elementary statements: Typically refer to objects, their properties and relations. But these are not explicitly represented in the propositional logic

Example:



Objects and properties are hidden in the statement, it is not possible to reason about them

(1) Statements that must be repeated for many objects

– In propositional logic these must be exhaustively enumerated

Example:

If John is a CS UPitt graduate then John has passed cs441

Translation:

John is a CS UPitt graduate -> John has passed CS441

Similar statements can be written for other Upitt graduates:

Ann is a CS Upitt graduate-> Ann has passed CS441

Ken is a CS Upitt graduate -> Ken has passed CS441

• What is a more natural solution to express the above knowledge?

(1) Statements that define the property of the group of objects

Solution:

make statements with quantifiers

- Universal quantifier: the property is satisfied by all members of the group
- Existential quantifier: at least one member of the group satisfies the property

Solution:

make statements with variables

- If x is a CS Upitt graduate then x has passed cs441
- x is a CS UPitt graduate -> x has passed cs441

(1) Statements that define the property of the group of objects

2.1.3 Predicate logic

Remedies the limitations of propositional logic

- Explicitly models objects and their properties
- Allows to make statements with variables and quantify them

Basic building blocks of the predicate logic:

- Constant –models a specific object. Examples: "John", "France", "7"
- Variable – represents an object of a specific type (defined by the universe of discourse)

Examples: x, y

- (universe of discourse can be people, students, numbers)
- Predicate - over one, two or many variables or constants. – Represents properties or relations among objects

Examples: $\text{Red}(\text{car23})$, $\text{student}(x)$, $\text{married}(\text{John}, \text{Ann})$

2.1.4 Predicates

A symbol which represents a property or relation. Often, the statement or relation itself is called a predicate.

Predicates are typically capital letters P, Q, R, S . Predicates are very much like propositional variables in that they stand in for something. Predicates are more general in that they need not have a truth value. Moreover, whereas propositions needed to be a declarative statement (i.e. an independent clause in a sentence), predicates can be an incomplete phrase.

Predicates represent properties or relations among objects. A predicate $P(x)$ assigns a value true or false to each x depend

A predicate $P(x)$ assigns a value true or false to each x depending on whether the property holds or not for x .

The assignment is best viewed as a big table with the variable x substituted for objects from the universe of discourse

The following are all predicates.

- $D := \text{"is a dog"}$

- $R :=$ "has rowing as their favourite sport"
- $P :=$ "is a positive number"
- $E :=$ "is an even integer"

Notice from these examples that predicates typically lack subjects. Combining predicates with variables allows us to form propositional functions.

Example:

- Assume $\text{Student}(x)$ where the universe of discourse are people
- $\text{Student}(\text{John}) \dots T$ (if John is a student)
- $\text{Student}(\text{Ann}) \dots T$ (if Ann is a student)
- $\text{Student}(\text{Jane}) \dots F$ (if Jane is not a student)
- • ...

Now,

Assume a predicate $P(x)$ that represents the statement:

What are the truth values of:

- | | |
|----------|---|
| • $P(2)$ | T |
| • $P(3)$ | T |
| • $P(4)$ | F |
| • $P(5)$ | T |
| • $P(6)$ | F |
| • $P(7)$ | T |

Also,

All statements $P(2), P(3), P(4), P(5), P(6), P(7)$ are propositions

What are the truth values of:

- | | |
|----------|---|
| • $P(2)$ | T |
| • $P(3)$ | T |
| • $P(4)$ | F |
| • $P(5)$ | T |
| • $P(6)$ | F |
| • $P(7)$ | T |

Assume a predicate $P(x)$ that represents the statement:

x is a prime number

Is $P(x)$ a proposition? No. Many possible substitutions are possible.

Also,

Predicates can have more arguments which represent the relations between objects

Example:

- Older(John, Peter) denotes 'John is older than Peter' – this is a proposition because it is either true or false
- Older(x,y) - 'x is older than y' – not a proposition, but after the substitution, it becomes one
- Predicates can have more arguments, which represent the relations between objects

Note: Predicates can have more arguments, which represent the relations between objects

Example:

- Let $Q(x,y)$ denote ' $x+5 > y$ '

Is $Q(x,y)$ a proposition? No!

Is $Q(3,7)$ a proposition? Yes. It is true.

What is the truth value of:

$Q(3,7)$ T

$Q(1,6)$ F

$Q(2,2)$ T

Is $Q(3,y)$ a proposition? No! We cannot say if it is true or false.

Note: Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value

Example

Let $Q(x, y)$ denote the statement " $x = y + 3$." What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Solution

To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$. Hence, $Q(1, 2)$ is the statement " $1 = 2 + 3$," which is false. The statement $Q(3, 0)$ is the proposition " $3 = 0 + 3$," which is true

In-Text Questions (ITQs)

1. What are the truth values of the propositions $R(1, 2, 3)$ and $R(0, 0, 1)$
2. Consider the statement if $x > 0$ then $x := x + 1$

In-Text Answers (ITAs)

1. The proposition $R(1, 2, 3)$ is obtained by setting $x = 1$, $y = 2$, and $z = 3$ in the statement $R(x, y, z)$. We see that $R(1, 2, 3)$ is the statement " $1 + 2$ "

In general, a statement involving the n variables $x_1, x_2, \dots, x_n$ can be denoted by

$sP(x_1, x_2, \dots, x_n)$.

A statement of the form $P(x_1, x_2, \dots, x_n)$ is the value of the propositional function P at the n -tuple $(x_1, x_2, \dots, x_n)$, and P is also called an n -place predicate or an n -ary predicate. Propositional functions occur in computer programs, as Example 6 demonstrates

2. When this statement is encountered in a program, the value of the variable x at that point in the execution of the program is inserted into $P(x)$, which is " $x > 0$." If $P(x)$ is true for this value of x , the assignment statement $x := x + 1$ is executed, so the value of x is increased by 1. If $P(x)$ is false for this value of x , the assignment statement is not executed, so the value of x is not changed

2.1.5 Quantifiers

Quantifiers are used as modifiers to variables, giving rise to the logical formalization of the English words "all", "some", "any", "every", etc. For example, "all men are mortal" or "some birds cannot fly" cannot be expressed in propositional logic but can be represented in predicate logic using quantifiers.

Quantifiers are words that refer to quantities such as "all" or "some" and they tell for how many elements a given predicate is true

When the variables in a propositional function are assigned values, the resulting statement becomes a proposition with a certain truth value. However, there is another important way, called quantification, to create a proposition from a propositional function. Quantification expresses the extent to which a predicate is true over

a range of elements. In English, the words all, some, many, none, and few are used in quantifications. We will focus on two types of quantification here:

We need quantifiers to formally express the meaning of the words "all" and "some".

The two main quantifiers are the universal quantifier and the existential quantifier.

Assign a value to x in $P(x)$ = "x is an odd number", so the resulting statement becomes a proposition:

$P(7)$ is true, $P(2)$ is false.

Introduced into logic by logicians Charles Sanders Pierce and Gottlob Frege during late 19th century Two types of quantifiers: 1. Universal quantifier ($\forall$) 2. Existential quantifier ($\exists$)

Universal quantifier

Definition

"For all", "For any", "Given any". A universal quantifier asserts that a predicate is true for any and every value in the domain. It is denoted $\forall$.

Universal quantification is a quantifier (something that tells the amount or quantity) meaning "given any" or "for all".

- $\forall$ universal quantifiers, "for all", "for every"
- $\in$ - "is a member (or) element of", "belonging to"
- **D** – domain of predicate variable

Let $p(x)$ be a predicate and D be the domain of x A universal statement is a statement of the form

$$\forall x \in D, p(x)$$

...

Forms:

" $p(x)$ is true for all values of x "

"For all x , $p(x)$ "

"For each x , $p(x)$ "

"For every x , $p(x)$ "

"Given any x , $p(x)$ " It is true if $p(x)$ is true for each x in D ;

It is false if $p(x)$ is false for at least one x in D

A counterexample to the universal statement is the value of x for which $p(x)$ is false

Example

Universal st.s	Domain	Truth value	Method
$\forall x \in D, x^2 \geq x$	$D = \{1, 2, 3\}$	True	Method of exhaustion
$\forall x \in \mathbb{R}, x^2 \geq x$	$\mathbb{R}$	False	Counterexample $x = 0.1$

Caution: Method of exhaustion cannot be used to prove universal statements for infinite sets

Existential quantifier

Definition

"There exists", "For some", "There is a", "There is at least one". An existential quantifier asserts that a predicate is true for at least one value in the domain. It is denoted $\exists$.

An existential quantification is a quantifier (something that tells the amount or quantity) meaning "there exists", "there is at least one" or "for some".

The existential quantification of $P(x)$ is the statement:

"There exists an element x in the domain such that $P(x)$ " denoted $\exists xP(x)$.

$\exists xP(x)$ is true when $P(x)$ is true for one or more x in the domain.

An element for which $P(x)$ is true is called a witness of $\exists xP(x)$.

$\exists xP(x)$ is false when $P(x)$ is false for every x in the domain (if domain nonempty).

If the domain is empty, $\exists xP(x)$ is false for any propositional function $P(x)$, since there are no witnesses in the domain.

If the domain is finite $\{x_1, x_2, \dots, x_n\}$, $\exists xP(x)$ is the same as $P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$.

That is

Many mathematical statements assert that there is an element with a certain property. Such statements are expressed using existential quantification. With existential quantification, we form a proposition that is true if and only if $P(x)$ is true for at least one value of x in the domain.

The existential quantification of $P(x)$ is the proposition

"There exists an element x in the domain such that $P(x)$."

We use the notation $\exists x P(x)$ for the existential quantification of $P(x)$.

Here $\exists$ is called the existential quantifier.

Note: $\exists x P(x)$ can be read as "There exists an such that" $P(x)$.

Let $P(x) := x > 0$. If U is all integers, $\exists x P(x)$ is *true*.

Let $P(x) := 2x$ is odd. If U is all integers, $\exists x P(x)$ is *false*.

Let $P(x) := x$ has feathers. If U is all animals, $\exists x P(x)$ is *true*.

Let $P(x) := x$ has feathers. If U is all birds, $\exists x P(x)$ is *true*.

Example:

- Let $T(x)$ denote $x > 5$ and x is from Real numbers.
- What is the truth value of $\exists x T(x)$?
- Answer:
- Since $10 > 5$ is true.

Therefore, it is true that $\exists x T(x)$

Example

Let $P(x)$ denote the statement " $x > 3$." What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Solution

Because " $x > 3$ " is sometimes true—for instance, when $x = 4$ —the existential quantification of $P(x)$, which is $\exists x P(x)$, is true.

Observe that the statement $\exists x P(x)$ is false if and only if there is no element x in the domain for which $P(x)$ is true. That is, $\exists x P(x)$ is false if and only if $P(x)$ is false for every element of the domain.

Example

What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement " $x^2 > 10$ " and the universe of discourse consists of the positive integers not exceeding 4?

Solution

Because the domain is $\{1, 2, 3, 4\}$, the proposition $\exists xP(x)$ is the same as the disjunction

$$P(1) \vee P(2) \vee P(3) \vee P(4).$$

Because $P(4)$, which is the statement " $42 > 10$," is true, it follows that $\exists xP(x)$ is true.

2.1.6 Other Vitals about Predicate Logic

Formal and informal languages

$$\forall x \in \mathbb{R}, x^2 \geq 0$$

Then

Every real number has a nonnegative square
 All real numbers have nonnegative squares
 Any real number has a nonnegative square
 The square of each real number is nonnegative
 No real numbers have negative squares
 x^2 is nonnegative for every real x
 x^2 is not less than zero for each real number x

Universal conditional statement ($\forall, \rightarrow$)

A universal conditional statement is of the form $\forall x$, if $p(x)$ then $q(x)$

Example

$\forall x \in \mathbb{R}$, if $x > 2$ then $x^2 > 4$

$\forall$ real number x , if x is an integer then x is rational

$\forall$ integer x , x is rational $\gg$ Logically equivalent

$\forall x$, if x is a square then x is a rectangle

$\forall$ square x , x is a rectangle $\gg$ Logically equivalent

$\forall x \in U$, if $p(x)$ then $q(x)$

$\forall x \in D$, $q(x)$ $\gg$ Logically equivalent

(where, $D = \{x \in U \mid p(x) \text{ is true}\}$)

Implicit quantification ($\Rightarrow, \Leftrightarrow$)

Example

If a number is an integer, then it is a rational number Implicit meaning: $\forall$ number x , if x is an integer, x is rational The number 10 can be written as a sum of two prime numbers Implicit meaning: $\exists$ prime numbers p and q such that $10 = p+q$ If $x > 2$, then $x^2 > 4$ Implicit meaning: $\forall$ real x , if $x > 2$, then $x^2 > 4$

Definition

Let $p(x)$ and $q(x)$ be predicates and D be the common domain of x .

Then implicit quant. symbols $\Rightarrow, \Leftrightarrow$ are defined as:

$$p(x) \Rightarrow q(x) \equiv \forall x, p(x) \rightarrow q(x)$$

$$p(x) \Leftrightarrow q(x) \equiv \forall x, p(x) \leftrightarrow q(x)$$

Example

$q(n)$: n is a factor of 8;

$r(n)$: n is a factor of 4

$s(n)$: $n < 5$ and $n \neq 3$ Domain of n is $\mathbb{Z}^+$ (i.e., positive integers)

What are the relationships between $q(n)$, $r(n)$, and $s(n)$ using symbols $\Rightarrow$ and $\Leftrightarrow$?

Solution

Truth set of $q(n) = \{1, 2, 4, 8\}$; Truth set of $r(n) = \{1, 2, 4\}$;

Truth set of $s(n) = \{1, 2, 4\}$

$$\forall n \text{ in } \mathbb{Z}^+, r(n) \rightarrow q(n)$$

i.e., $r(n) \Rightarrow q(n)$ i.e., " n is a factor of 4" $\Rightarrow$ " n is a factor of 8"

$$\forall n \text{ in } \mathbb{Z}^+, r(n) \leftrightarrow s(n)$$

i.e., $r(n) \Leftrightarrow s(n)$ i.e., " n is a factor of 4" $\Leftrightarrow$ " $n < 5$ and $n \neq 3$ "

$$\forall n \text{ in } \mathbb{Z}^+, s(n) \rightarrow q(n) \text{ i.e., } s(n) \Rightarrow q(n)$$

i.e., " $n < 5$ and $n \neq 3$ " $\Rightarrow$ " n is a factor of 8"

Relation between quantifiers ($\forall$, $\exists$) and ($\wedge$, $\vee$)

Universal statements are generalizations of and statements

Existential statements are generalizations of or statements

If $p(x)$ is a predicate and $D = \{x_1, x_2, \dots, x_n\}$ is the domain of x , then

Important

Without specifying a domain of discourse, a quantifier applied to a propositional function cannot give us a truth value. The predicate, the domain, and choice of quantifier together give a truth value.

Quantifiers and domains

Let $P(x) := x < 2$.

1. If U is the positive integers, what is the truth value of $\forall x P(x)$?
? What is the truth value of $\exists x P(x)$?
2. If U is the negative integers, what is the truth value of $\forall x P(x)$?
What is the truth value of $\exists x P(x)$?
3. If U is the numbers 3, 4, and 5, what is the truth value of $\forall x P(x)$?
What is the truth value of $\exists x P(x)$?

Solution

1. $\forall x P(x)$ is **false**. $\exists x P(x)$ is **true**.
2. $\forall x P(x)$ is **true**. $\exists x P(x)$ is **true**.
3. $\forall x P(x)$ is **false**. $\exists x P(x)$ is **false**.

The above examples suggest implications between existential and universal quantifiers.

Let $P(x)$ be a propositional function with a universe U .

If $\forall x P(x)$ is true, then $\exists x P(x)$ *must* be true.

If $\exists x P(x)$ is false, then $\forall x P(x)$ *must* be false.

That is, $\forall x P(x) \rightarrow \exists x P(x)$ and

$\neg(\exists x P(x)) \rightarrow \neg(\forall x P(x))$.

For some domains of discourse, it is possible to “expand” a quantified expression into an equivalent compound expression with no variables. In particular, when domains are finite.

Finite domains, conjunctions, disjunctions

Let P be a predicate and U be the integers 1, 2, and 3.

$$\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$$

$$\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$$

Precedence of Quantifiers

Quantifiers have a higher precedence than all other logical connectives.

$$\forall x P(x) \vee Q(x) \equiv (\forall x P(x)) \vee Q(x)$$

$$\exists x P(x) \wedge Q(x) \equiv (\exists x P(x)) \wedge Q(x)$$

$$\forall x P(x) \rightarrow Q(x) \equiv (\forall x P(x)) \rightarrow Q(x)$$

$$P(x) \rightarrow \exists x Q(x) \equiv P(x) \rightarrow (\exists x Q(x))$$

2.1.7 Conclusion

Predicate logic enables deeper expression of mathematical and real-world relationships through variables and quantifiers. Understanding its structure and application lays the groundwork for advanced topics such as first-order logic, formal verification, and database query languages. Mastery of this unit is crucial for logical reasoning in both theoretical and applied computing contexts.

Summary

This unit introduced Predicate Logic, which extends propositional logic by including variables, predicates, and quantifiers. Predicates express properties or relationships between objects and become propositions when specific values are substituted. Quantifiers universal ($\forall$) and existential ($\exists$) allow reasoning over entire domains. Through examples and truth table analysis, the unit emphasises the use of predicate

logic in constructing precise mathematical and logical statements relevant to computer science and formal reasoning

Self-Assessment Questions

1. What is the difference between a proposition and a non-propositional statement? Provide two examples of each.
2. Construct a truth table for the compound statement: $\neg(p \wedge q) \rightarrow (r \vee p)$.
3. How does the implication $p \rightarrow q$ differ from the biconditional $p \leftrightarrow q$ in terms of truth values?

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. Let $P(x, y)$ denote $x + y = 17$, and let U be the set of integers. Show that
 - (a) $\forall x (\exists y (P(x, y)))$ is true.
 - (b) $\exists y (\forall x (P(x, y)))$ is false
2. Find formulas in negation normal form equivalent to each of the following formulas. (In cases (c) and (d), the intended universal set is the set of all real numbers, but that does not affect the answers.)
 - (a) $\neg \forall x \in \mathbb{N} (x \text{ is prime} \rightarrow 2 + 1 \text{ is even})$
 - (b) $\neg \exists x \in \mathbb{Q} (x > 0 \wedge x^3 = 2)$
 - (c) $\neg \exists x (\forall y (xy = y))$
 - (d) $\neg \forall x (\forall y (x < y \rightarrow (\exists z (x < z \wedge z < y))))$

References/Further Readings

- Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.
- Epp, S. S. (2011). Discrete Mathematics with Applications (4th ed.). Cengage Learning.
- Enderton, H. B. (2001). A Mathematical Introduction to Logic (2nd ed.). Academic Press

Module Three

Set

Unit 1 — Set

INTRODUCTION

A set is one of the most fundamental objects in mathematics. A set is an unordered collection of objects. A set is a collection of elements. A set is simply a collection of distinct objects, called elements or members, grouped together because they share some property or characteristic.

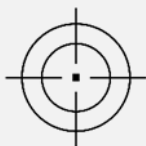
For Example

The set {white, red, green} contains the names of the colors white, red, and green and nothing else.

The set {0, 1, 2, 3, 4, 5, 1003456792311} contains seven integers.

The set {red, yellow, blue} contains the names of primary colors.

A set of stamps stored in loose-leaf notebooks on a shelf is usually called a stamp collection.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Define and identify various types of sets, including empty, singleton, finite, infinite, equal, and equivalent sets.
- Apply set-theoretic notation and understand set membership using symbols like $\in$, $\notin$, $\subseteq$, $\subset$, and $P(A)$.
- Perform basic and advanced set operations such as union, intersection, difference, symmetric difference, and complement, with appropriate use of Venn diagrams.
- Describe relationships between sets and determine subset, proper subset, and disjoint set conditions with examples

3.1.1 Basic Characteristics of a Set

For any set A and any element b , either b is in A or b is not in A . If you ask whether an element is in a set, the answer is either yes or no

Is 0 in $\{1, 21\}$ No.

Is 0 in $\{0, 1, 2, 3, 4, 5, 1003456792311\}$ Yes.

Is New York in $\{\text{Liverpool, London, Los Angeles}\}$ No.

Is green in $\{\text{red, yellow, blue}\}$ No.

Is New York in $\{\text{England, France, United States}\}$? No.

In mathematical terminology,

0 is an element of $\{0, 1, 2, 3, 4, 5, 1003456792311\}$, and green is not an element of $\{\text{red, yellow, blue}\}$.

The expression "is an element of" is denoted by the symbol $\in$, a form of the Greek letter epsilon. For example, we write

$0 \in \{0, 1, 2, 3, 4, 5, 1003456792311\}$

And

$\text{green} \notin \{\text{red, blue, yellow}\}$

The slash through the $\notin$ symbol means not, just as it does in $\neq$. Is a member of, is contained in, or simply is in means the same as "is an element of." Mathematics, like ordinary language, is full of synonyms.

Examples

- In a class of 13 students, at least two must be born in the same month. Here, the 13 students are "pigeons" and the 12 months are "pigeonholes".
- If 102 students took an exam with maximal score 100 points, then at least two students will have the same score. The students are "pigeons", the numbers of points are "pigeonholes".
- In a letter with 30 words at least two words begin with the same letter. The words are "pigeons" and the 26 letters are "pigeonholes".
- Among 100 integers $a_1, \dots, a_{100}$ one can find two $a_i, a_j, i \neq j$, whose difference is divisible by 97. Integers $a_1, \dots, a_{100}$ are "pigeons", residues mod 97 are "pigeonholes".

- A drawer contains 10 pairs of socks of different colors and you pick some randomly. What minimum number guarantees a pair of one color? The 10 colors are "pigeonholes", so we need to pick 11 to guarantee.

Pigeonhole Principle: If k is a positive integer and $k + 1$ objects are placed into k boxes, then at least one of the boxes will contain two or more objects

Set-Theoretic Notation

Three methods to describe the set with elements 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9 are:

1. List the elements in braces: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}.

We can also abbreviate this list as {0, 1, 2,..., 9}.

2. Describe the elements in terms of some property they satisfy:

$\{x : x \text{ is an integer and } x > -1/2 \text{ and } x < 19/2\}$

This notation is read as "the set of (all) x such that x is an integer and x is greater than minus one-half, and x is less than nineteen halves."

The colon is read as "such that." The description following the colon tells what property these x 's have.

3. Describe the elements as the set of all elements in some other set that satisfy some property.

Here, if Z denotes the set of integers, then the set can be defined as

$\{x \in Z : x > -1/2 \text{ and } x < 19/2\}$

We often list the elements of a set in a way that shows an obvious association between the natural numbers and the elements of the set. For example, $1, 2, 2^2, 2^3, 2^4, \dots$

We call such a set a sequence. We can refer to a sequence by $a_0, a_1, a_2, \dots$. In the above example, we have $a_0 = 1, a_1 = 2, a_2 = 2^2, \dots, a_n = 2^n, \dots$

Special Sets

There are special names for certain common sets of numbers. Some of them are listed here.

Special Sets

$\mathbb{N}$: the set of **natural numbers**, or the set of non-negative integers {0, 1, 2, 3, 4, ...}.

$\mathbb{Z}$: the set of **integers**, or {..., -3, -2, -1, 0, 1, 2, 3, ...}.

$\mathbb{Q}$: the set of **rational numbers**, or the set of fractions of integers with nonzero denominator, such as $\frac{1}{3}$ or $\frac{2357}{9731}$.

$\mathbb{R}$: the set of **real numbers**, or the set of numbers written with a decimal point, such as $\pi = 3.14159\dots$, -2.715 , or $2.35353535\dots$

$\emptyset$: the **empty set**, or the set { } with no elements.

In some circumstances, it is convenient to write the set of squares of natural numbers as rather than as and for some

Set Membership

To prove an element is a member of a set, you must prove that the element shares the property that defines membership. For example, we can define the notion of a number being a prime without knowing that any particular number is a prime. We must then show that any number we think is a prime has the defining property. First, we need to know what a divisor is before we can define a prime. For integers m and n , we say m is a divisor of n , denoted as $m|n$, if there is a natural number k such that $n = m.k$. A natural number p is prime if $p \neq 1$ and its only divisors are 1 and p . Let $P = \{n : n \in \mathbb{N} \text{ and } n \text{ is a prime}\}$. In Example below, we will show that P is nonempty.

Example. Prove that 3 is a prime—that is, that $3 \in P$.

Solution.

We must show 3 has the property that its only divisors are 1 and 3. Since the only other possibility is 2 and 2 does not divide 3, 3 is therefore a prime. A divisor of an integer is also called a factor.

3.1.2 Types of Sets in Mathematics

Let's explore the different types of sets along with definitions and examples.

Singleton Set

Singleton Sets are those sets that have only 1 element present in them.

Example:

- Set $A = \{1\}$ is a singleton set as it has only one element, that is, 1.
- Properties of Singleton Set:
 - A singleton set has exactly one element, for example: $\{7\}$
 - Its cardinality (size) is 1.
 - A singleton set is always a finite set.
 - It is a subset of any set that contains its element, for example: $\{7\} \subseteq \{5, 6, 7\}$
 - The power set of a singleton set has 2 elements:
 - Example: Power set of $\{a\} = \{\emptyset, \{a\}\}$
 - Every element of a singleton set is also a singleton set in the power set.
- Set $P = \{a : a \text{ is an even prime number}\}$ is a singleton set as it has only one element 2.

Similarly, all the sets that contain only one element are known as Singleton sets.

Empty Set

Empty sets are also known as Null sets or Void sets. They are the sets with no element/elements in them. They are denoted as $\emptyset$ also known as phi.

Example:

- Set A = {a: a is a number greater than 5 and less than 3}
- Set B = {p: p are the students studying in class 7 and class 8}

Properties of Empty Set ($\emptyset$):

- $\emptyset$ has no elements (0 elements).
- $\emptyset$ is a subset of every set.
- There is only one empty set.
- $\emptyset$ is not a singleton set.
- The intersection of disjoint sets is $\emptyset$.
- The power set of $\emptyset$ is $\{\emptyset\}$ (has 1 element).

Finite Set

A finite set is a set that has a countable number of elements.

In other words, you can count how many elements it has, and the counting ends. They will be called a Finite set.

Example:

- A = {1, 2, 3, 4} → Finite (4 elements)
- B = {"apple", "banana"} → Finite (2 elements)
- C = $\emptyset$ (Empty set) → Also a finite set (0 elements)

Properties of Finite Set:

- Has a fixed number of elements.
- The number of elements is a whole number (0 or more).
- You can count the elements and finish counting.
- The empty set is also finite (with 0 elements).
- All singleton sets are finite.
- The union, intersection, or difference of two finite sets is also a finite set.

Infinite Set

Infinite Sets are those that have an infinite number of elements present, cases in which the number of elements is hard to determine are known as infinite sets.

Example:

- $A = \{1, 2, 3, 4, 5, 6, \dots\} \rightarrow$ Infinite
- $B =$ Set of all stars in the sky $\rightarrow$ Infinite
- $C =$ Set of all natural numbers $\rightarrow$ Infinite

Properties of Infinite Set:

- Has unlimited or endless elements.
- You cannot count all the elements — counting never ends.
- Removing or adding a finite number of elements doesn't make it finite.
- Infinite sets can be countable (like natural numbers) or uncountable (like real numbers).
- The power set of an infinite set is also infinite.

Equal Set

Two sets having the same elements and an equal number of elements are called equal sets. The elements in the set may be rearranged, or they may be repeated, but they will still be equal sets.

Example:

- Set $A = \{1, 2, 6, 5\}$
- Set $B = \{2, 1, 5, 6\}$

In the above example, the elements are 1, 2, 5, 6. Therefore, $A = B$.

Properties of Equal Sets:

- Two sets are equal if every element of one set is also in the other set.
- The order of the elements doesn't matter.
- The repetition of elements doesn't matter (sets don't allow duplicates).
- If $A = B$, then $A \subseteq B$ and $B \subseteq A$.
- Equal sets have the same number of elements (same cardinality).

Equivalent Set

Equivalent Sets are those which have the same number of elements present in them. It is important to note that the elements may be different in both sets but the number of elements present is equal. For Instance, if a set has 6 elements in it, and the other set also has 6 elements present, they are equivalent sets.

Example:

- Set A = {2, 3, 5, 7, 11}
- Set B = {p, q, r, s, t}
- Set A and Set B both have 5 elements; hence, both are equivalent sets.

Properties of Equivalent Sets:

- Equivalent sets have the same number of elements.
- The elements can be different; it is only the count that matters.
- If two sets A and B are equivalent, we write: $A \approx B$
- All equal sets are equivalent, but not all equivalent sets are equal.

Subset

Set A will be called the Subset of Set B if all the elements present in Set A already belong to Set B. The symbol used for the subset is $\subseteq$.

If A is a Subset of B, it will be written as $A \subseteq B$

Example:

- Set A = {33, 66, 99}
- Set B = {22, 11, 33, 99, 66}

Then, Set $A \subseteq$ Set B

Properties of Subsets:

- $A \subseteq A$ (Every set is a subset of itself)
- $\emptyset \subseteq A$ (Empty set is a subset of every set)
- If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$ (transitive property)
- If $A \subseteq B$ and $A \neq B$, then A is a proper subset of B (written $A \subset B$)
- A set with n elements has 2^n subsets (including $\emptyset$ the set itself)

Proper Subset

A proper subset is a subset that contains some but not all elements of the original set. It is not equal to the original set. If A and B are sets, and A is a proper subset of B, we write: $A \subset B$

Example:

Let $B = \{1, 2, 3\}$

- Proper subsets of B are:
- $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$
- $\{1, 2, 3\}$ is a subset of B, but not a proper subset, because it is equal to B.

Properties of a Proper Subset:

- A proper subset has fewer elements than the original set.
- Every proper subset is a subset, but not every subset is proper.
- The empty set ($\emptyset$) is a proper subset of any non-empty set.
- The original set is not a proper subset of itself.

Power Set

The set of any set A is defined as the set containing all the subsets of set A. It is denoted by the symbol $P(A)$ and read as the any Power set of A.

For any set A containing n elements, the total number of subsets formed is 2^n . Thus, the power set of A, $P(A)$, has 2^n elements.

It includes:

- The empty set ($\emptyset$)
- All individual elements as sets
- All combinations of elements
- The set itself

Example: For any set $A = \{a, b, c\}$, the power set of A is?

Solution:

Power Set $P(A)$ is,

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}$$

Properties of Power Set :

- If a set has n elements, its power set has 2^n subsets.
Example: $A = \{a, b, c\} \rightarrow 3 \text{ elements} \rightarrow \text{power set has } 2^3 = 8 \text{ subsets}$
- The power set always includes:
 - The empty set ($\emptyset$)
 - The original set
- Power set of $\emptyset$ is: $P(\emptyset) = \{\emptyset\}$

- The power set is always larger than the original set (except when the set is empty).

Universal Set

A universal set is a set that contains all the elements of the rest of the sets. It can be said that all the sets are subsets of the universal set. The universal set is denoted as U .

Example: For Set $A = \{a, b, c, d\}$ and Set $B = \{1, 2\}$, find the universal set containing both sets.

Solution:

Universal Set U is,

$$U = \{a, b, c, d, e, 1, 2\}$$

Properties of Universal Set:

- Contains all the elements being discussed.
- Every set is a subset of the universal set.
- The complement of a set is taken to be disjoint from the universal set.
- It is usually finite or defined by context (can be infinite in some cases).
- The intersection of U with any set A is A :
- $U \cap A = A$
- The union of U with any set A is U :
- $U \cup A = U$

Disjoint Sets

Two sets A and B that have no common elements are called Disjoint Sets. The intersection of the Disjoint set is ϕ , now for set A and set B $A \cap B = \phi$.

Example: Check whether Set $A = \{a, b, c, d\}$ and Set $B = \{1, 2\}$ are disjoint or not.

Solution:

$$\text{Set } A = \{a, b, c, d\}$$

$$\text{Set } B = \{1, 2\}$$

$$\text{Here, } A \cap B = \phi$$

Thus, Set A and Set B are disjoint sets.

Properties of Disjoint Sets:

- No common elements $\rightarrow A \cap B = \emptyset$
- If sets are disjoint, their intersection is the empty set.
- Disjoint sets can be finite or infinite.
- Two sets can be disjoint even if one or both are empty.
- In Venn diagrams, disjoint sets do not overlap.

3.1.3 Set Operations

Set Operations can be defined as the operations performed on two or more sets to obtain a single set containing a combination of elements from all the sets being operated upon.

- Set operations are mathematical operations operated on sets, which are collections of distinct objects or elements.
- There are three major types of operation on sets: Union ($\cup$), Intersection ($\cap$), and Difference ($-$).

Other operations include Difference, Complement, Addition, & Subtraction.

The most common set operations, such as union, intersection, disjoint, set difference, etc., will be explored in detail below, including their definitions, examples, and Venn diagrams.

Note: A Venn Diagram is a great tool used to visualise and represent operations on sets. It is used to display sets as intersecting circles. We can highlight a region under consideration when we carry out an operation. The cardinality of a set can be represented by putting numbers in the corresponding area.]

Union of Sets

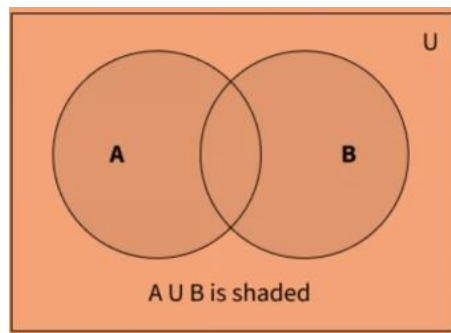
The **Union of sets** A and B, denoted by **$A \cup B$** , is the set of distinct elements that belong to set A or set B, or both. This operation can be represented as;

$$A \cup B = \{x: x \in A \text{ or } x \in B\}$$

In simple terms, the union includes every element that appears in either of the two sets, without any repetition. Where x is the elements present in both sets A and B.

Venn Diagram for the Union of Sets

The area shaded in green represents $A \cup B$ or the union of sets A and B.



Venn diagram of $A \cup B$

Example: Find the union of $A = \{2, 3, 4\}$ and $B = \{3, 4, 5\}$.

Solution:

$$A \cup B = \{2, 3, 4, 5\}.$$

Intersection

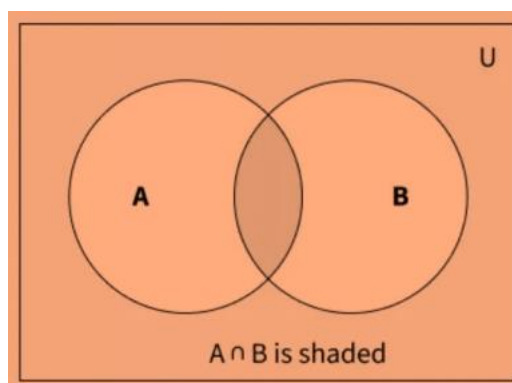
The *intersection of the sets* A and B, denoted by $A \cap B$, is the set of elements that belong to both A and B, i.e., the set of common elements in A and B. This operation is represented as:

$$A \cap B = \{x: x \in A \text{ and } x \in B\}$$

In other words, the intersection contains only those elements that are *present in both sets*. Here, x represents the elements that are common to both sets A and B.

Venn Diagram For Intersection of Sets

The area shaded in green represents $A \cap B$ or the intersection of sets A and B, which includes the elements common to both sets A and B.



Example: Find the intersection of $A = \{2, 3, 4\}$ and $B = \{3, 4, 5\}$

Solution:

Look for elements that are **common to both** sets A and B.

$$A \cap B = \{3, 4\}.$$

The intersection of sets A and B can also be interpreted as:

$$A \cap B = n(A) + n(B) - n(A \cup B)$$

Where,

$n(A)$ = the cardinality (or number of elements) of set A,

$n(B)$ = the cardinality of set B,

$n(A \cup B)$ = the cardinality of the union of sets A and B.

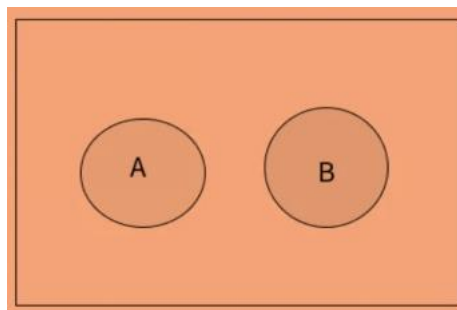
Disjoint Set

Two sets are said to be disjoint if their intersection is the empty set. i.e., sets have no common elements. In simpler terms, they don't "overlap" at all.

So if you try to find their **intersection**, you'll get the **empty set**, which we denote by the symbol ϕ or $\{\}$.

Venn Diagram For Disjoint Sets

The sets A and B are **disjoint**, meaning they have **no common elements** (no overlap).



Venn Diagram of A Disjoint B

For Example: Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8\}$

Solution:

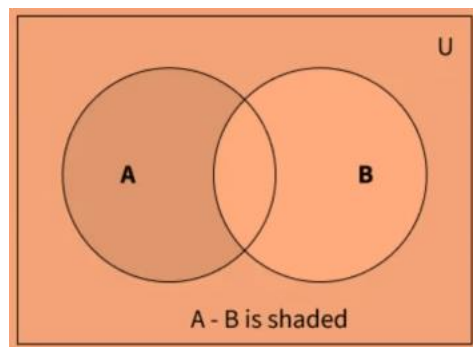
A and B are disjoint sets since both of them have no common elements.

Set Difference

The difference between sets is denoted by '**A - B**', which is the set containing elements that are in A but not in B i.e., all elements of A except the element of B.

Venn Diagram For Set Difference

In the below diagram, the set difference $A - B$ contains all the elements that are in A but not in B .



Example: If $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 6, 8\}$, find $A - B$.

Solution:

$$A - B = \{1, 3, 5\}$$

We can also state that the difference between set A and set B is equivalent to the intersection of set A with the complement of set B . Therefore,

$$A - B = A \cap B'$$

where $B' =$ the complement of set B .

Symmetric Difference

The symmetric difference of A and B includes elements in **A or B** but **not both**.

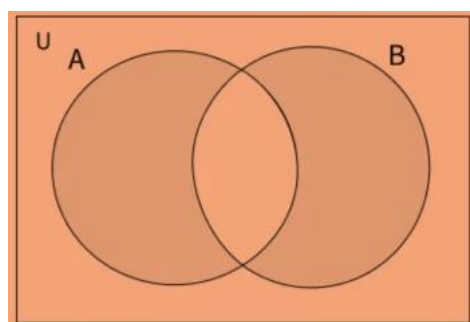
It is denoted by: **$A \Delta B$ or $A \oplus B$** .

The symmetric difference is like saying, "Give me everything that's not shared."

It is defined as: $A \Delta B = (A - B) \cup (B - A)$

Venn Diagram For Symmetric Difference

The symmetric difference $A \Delta B$ includes elements that are in either A or B but not in both.



Example: Let set $A = \{1, 2, 3\}$, and set $B = \{3, 4, 5\}$, then Find the symmetric difference

Solution:

$$A \triangle B = \{1, 2, 4, 5\}$$

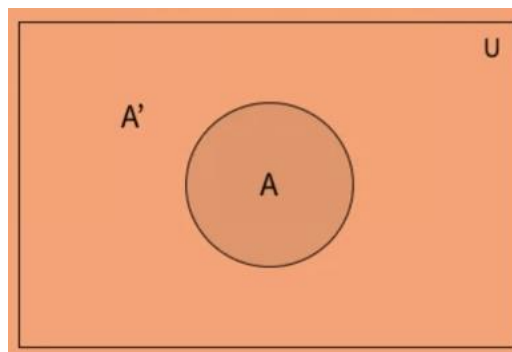
Complement of a Set

If U is a universal set and X is any subset of U , then the complement of X consists of all the elements in U that are not in X .

$$X' = \{a : a \in U \text{ and } x \notin A\}$$

Venn Diagram For Complement of a Set

In the diagram below, set A' includes all elements not in A , relative to the universal set.



Complement of a Set

Example: Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ And $A = \{1, 2, 5, 6\}$

Solution:

Then, the complement of A , denoted as A' , will be: $A' = \{3, 4, 7, 8\}$

In-Text Questions (ITQs)

If there are $n > 1$ people who can shake hands with one another, then there is always two persons who will shake hands with the same number of people

In-Text Answers (ITAs)

Each person shakes hands from 0 to $n - 1$ people, totally n possibilities. But 0 means that someone shakes hands to nobody, while $n - 1$ means shaking hands to everybody. So, both 0 and $n - 1$ cannot happen at the same time and this leaves n people to $n - 1$ possibilities.

3.1.4 Addition and Subtraction

The addition of sets A and B, known as Minkowski addition, results in a new set where each element is the sum of every possible pair of elements, one from set A and one from set B. In other words, the new set contains all possible sums of elements taken from the two sets. Set subtraction operates similarly but involves subtracting each element of set B from each element of set A. These operations are only meaningful when applied to numeric data types; if applied otherwise, they become symbolic with no practical significance.

Additionally, set addition is commutative, meaning the order of the sets does not affect the result. However, set subtraction is not commutative, as the order of the sets does affect the outcome.

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$A - B = A \cap B'$$

Example

For any choice of six digits in the set $S = \{1, 2, \dots, 9\}$ one can find two chosen digits giving in sum 10.

Solution: Pigeons here are digits and pigeonholes are 5 subsets $\{1, 9\}$, $\{2, 8\}$, $\{3, 7\}$, $\{4, 6\}$, $\{5\}$. Among six chosen digits two will be in the same subset, and thus, give in sum 10 Applying first-order differential

3.1.5 Conclusion

Understanding sets and their operations equips students with critical tools for reasoning, classification, and analysis in mathematics and computer science. Sets provide a simple yet powerful way to model and solve problems involving grouping, comparison, and logical structure.

Summary

The concept of sets is foundational in mathematics and serves as a basis for understanding collections of objects defined by shared characteristics. A set is a well-defined collection of distinct elements. The unit covered the basic definitions, notation, types of sets, and operations on sets using both symbolic and diagrammatic approaches. Special sets like the universal set, power set, and disjoint sets were explored. Lastly, fundamental set operations such as union, intersection, difference, and complement were illustrated using real-world and numeric examples.

Self-Assessment Questions

1. Define a universal set and give one example.
2. What is the power set of $A = \{x, y\}$? List all its elements.
3. Differentiate between equal sets and equivalent sets using your own examples.

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes))

Tutor-Marked Assignment

1. Given the sets $A = \{2, 4, 6, 8\}$, $B = \{3, 4, 5, 6\}$, and $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, perform the following operations:
 - a. $A \cup B$
 - b. $A \cap B$
 - c. $A - B$
 - d. B' (complement of B in U)

2.

- a. Prove that 7 is a prime number using the definition of divisors.
- b. Construct a Venn diagram for the following sets and shade the region representing $A \triangle B$:

$$A = \{1, 2, 3, 4\}, B = \{3, 4, 5, 6\}$$

References/Further Readings

Tucker, A. (2012). Applied Combinatorics. John Wiley & Sons.

Rosen, K. H. (2019). Discrete Mathematics and Its Applications (8th ed.). McGraw-Hill.

Grimaldi, R. P. (2004). Discrete and Combinatorial Mathematics: An Applied Introduction. Pearson Education.

Epp, S. S. (2010). Discrete Mathematics with Applications. Brooks/Cole Cengage Learning

Module Four

Function

Unit 1 — Functions

INTRODUCTION

Functions are incredibly important to the study of mathematics. We have all seen functions in some way during our previous mathematical education. Although functions are usually only introduced informally, and for the case of real numbers.

The fundamental part of a function is it takes some input and returns another



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Define a function and identify its domain, codomain, range, image, and pre-image.
- Differentiate between types of functions such as one-one, onto, many-one, into, and bijective.
- Analyse functions using algebraic notation, arrow diagrams, and tabular representations.

4.1.1 Function

A function is a rule that assigns each input exactly one output. We call the output the image of the input. The set of all inputs for a function is called the domain. The set of all allowable outputs is called the codomain. We would write $f:X \rightarrow Y$ to describe a function with name f , domain X and codomain Y .

For example, consider the function $f:N \rightarrow N$ defined by $f(x)=x^2+3$. Here the domain and codomain are the same set (the natural numbers). The rule is: take your input, multiply it by itself and add 3.

The key thing that makes a rule a function is that there is exactly one output for each input. That is, the rule must be good. What output do we assign to the input 7? There can only be one answer for any particular function.

A function f from a set A to a set B assigns each element of A to exactly one element of B .

A is called domain of f , and B is called codomain of f .

If f maps element $a \in A$ to element $b \in B$, we write $f(a) = b$

If $f(a) = b$, b is called image of a ; a is in preimage of b .

Range of f is the set of all images of elements in A .

Example

The following are all examples of functions:

1. $f:Z \rightarrow Z$ defined by $f(n)=3n$. The domain and codomain are both the set of integers. However, the range is only the set of integer multiples of 3.
2. $g:\{1,2,3\} \rightarrow \{a,b,c\}$ defined by $g(1)=c$, $g(2)=a$ and $g(3)=a$. The domain is the set $\{1,2,3\}$, the codomain is the set $\{a,b,c\}$ and the range is the set $\{a,c\}$. Note that $g(2)$ and $g(3)$ are the same element of the codomain. This is okay since each element in the domain still has only one output.
3. $h:\{1,2,3,4\} \rightarrow N$ defined by the table:

x	1	2	3	4
$h(x)$	3	6	9	12

Here the domain is the finite set $\{1,2,3,4\}$ and to codomain is the set of natural numbers, N . At first you might think this function is the same as f defined above. It is

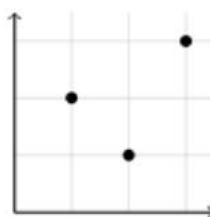
absolutely not. Even though the rule is the same, the domain and codomain are different, so these are two different functions.

Pre-Image and Image of a Function

A function $f: A \rightarrow B$ such that for each $a \in A$, there exists a unique $b \in B$ such that $(a, b) \in R$ then, a is called the pre-image of f and b is called the image of f .

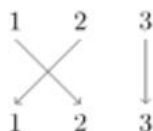
Describing Functions

Describing a function graphically usually means drawing the graph of the function: plotting the points on the plane. We can do this, and might get a graph like the following for a function $f: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$.



It would be absolutely **WRONG** to connect the dots or try to fit them to some curve. There are only three elements in the domain. A curve would mean that the domain contains an entire interval of real numbers.

Here is another way to represent that same function:



This shows that the function f sends 1 to 2, 2 to 1 and 3 to 3: just follow the arrows.

The arrow diagram used to define the function above can be very helpful in visualizing functions. We will often be working with functions with finite domains, so this kind of picture is often more useful than a traditional graph of a function.

Note that for finite domains, finding an algebraic formula that gives the output for any input is often impossible. Of course we could use a piecewise defined function, like'

$$f(x) = \begin{cases} x + 1 & \text{if } x = 1 \\ x - 1 & \text{if } x = 2 \\ x & \text{if } x = 3 \end{cases}$$

This describes exactly the same function as above, but we can all agree is a ridiculous way of doing so.

Since we will so often use functions with small domains and codomains, let's adopt some notation to describe them. All we need is some clear way of denoting the image of each element in the domain. In fact, writing a table of values would work perfectly:

x	0	1	2	3	4
$f(x)$	3	3	2	4	1

We simplify this further by writing this as a "matrix" with each input directly over its output:

$$f = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ 3 & 3 & 2 & 4 & 1 \end{pmatrix}.$$

Note this is just notation and not the same sort of matrix you would find in a linear algebra class (it does not make sense to do operations with these matrices, or row reduce them, for example).

One advantage of the two-line notation over the arrow diagrams is that it is harder to accidentally define a rule that is not a function using two-line notation

4.1.2 Types of Function

Some of the common types of functions are: One-One Function, Many-One Function, Onto Function, Into Function, One-One Correspondent Function, One-One Into Function, Many-One Onto Function, Many-One Into Function. Let's discuss these in detail.

One-One Function (or Injective Function):

A function in which one element of the domain is connected to one element of the codomain.

A function $f: A \rightarrow B$ is said to be a one-one (injective) function if different elements of A have different images in B .

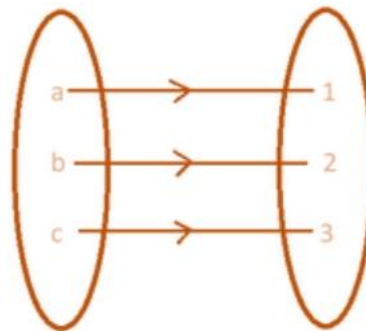
$f: A \rightarrow B$ is one-to-one

$$\Rightarrow a \neq b \Rightarrow f(a) \neq f(b) \quad \text{for all } a, b \in A$$

$$\Rightarrow f(a) = f(b) \Rightarrow a = b \quad \text{for all } a, b \in A$$

ONE-ONE FUNCTION

Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$ are two sets



Many-One Function

A function $f: A \rightarrow B$ is said to be a many-one function if two or more elements of set A have the same image in B.

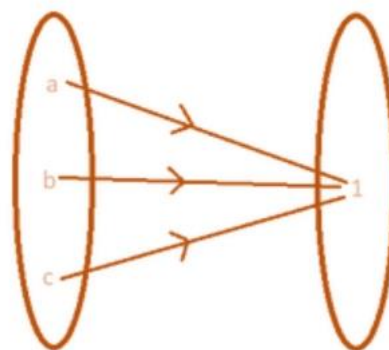
A function $f: A \rightarrow B$ is a many-one function if it is not a one-one function.

$f: A \rightarrow B$ is many-one.

$\Rightarrow a \neq b$ but $f(a) = f(b)$ for all $a, b \in A$

MANY-ONE FUNCTION

Let $A = \{a, b, c\}$ and $B = \{1\}$ are two sets



Onto Function (or Surjective Function)

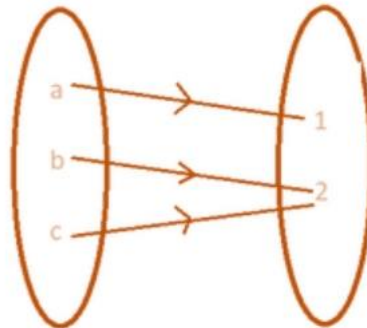
A function $f: A \rightarrow B$ is said to be onto (surjective) function if every element of B is an image of some element of A i.e. $f(A) = B$ or range of f is the codomain of f.

A function in which every element of the codomain has one pre-image.

$f: A \rightarrow B$ is onto if for each $b \in B$, there exists $a \in A$ such that $f(a) = b$.

ONTO FUNCTIONS

Let $A = \{a, b, c\}$ and $B = \{1, 2\}$ are two sets



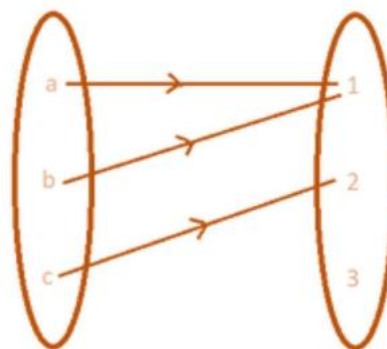
Into Function

A function $f: A \rightarrow B$ is said to be an into a function if there exists an element in B with no pre-image in A.

A function $f: A \rightarrow B$ is into function when it is not onto.

INTO FUNCTION

Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$ are two sets



One-One Correspondent Function (or Bijective Function or One-One Onto Function)

A function which is both one-one and onto (both injective and surjective) is called one-one correspondent(bijective) function.

$f: A \rightarrow B$ is one-one correspondent (bijective) if:

one-one i.e. $f(a) = f(b)$

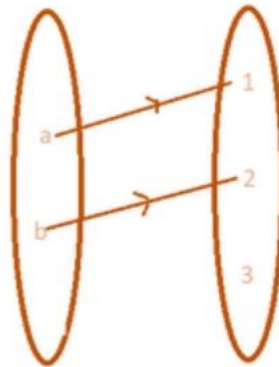
$\Rightarrow a = b$ for all $a, b \in A$

onto i.e. for each $b \in B$, there exists

$a \in A$ such that $f(a) = b$.

ONE-ONE INTO FUNCTION

Let $A = \{a, b\}$ and $B = \{1, 2, 3\}$ are two sets

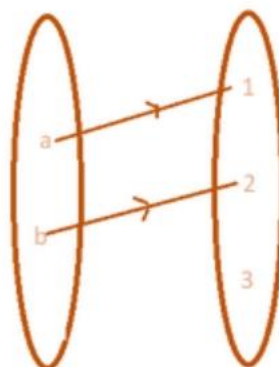


One-One Into Function

A function that is both one-one and into is called one-one into function.

ONE-ONE INTO FUNCTION

Let $A = \{a, b\}$ and $B = \{1, 2, 3\}$ are two sets

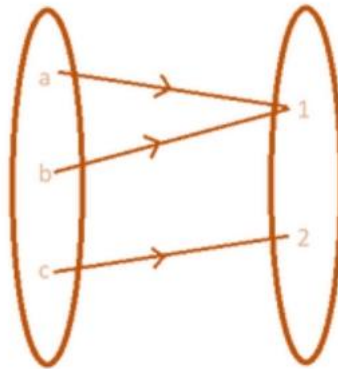


Many-One Onto Function

A function that is both many-one and onto is called many-one onto function.

MANY-ONE ONTO FUNCTION

Let $A = \{a, b, c\}$ and $B = \{1, 2\}$ are two sets

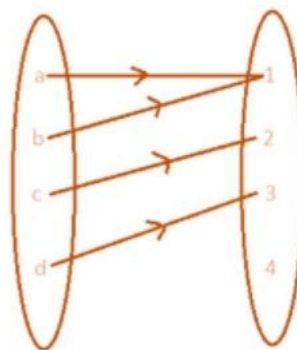


Many-One Into Function

A function that is both many-one and into is called many-one into function.

MANY-ONE INTO FUNCTION

Let $A = \{a, b, c, d\}$ and $B = \{1, 2, 3, 4\}$ are two sets



Inverse of a Function

Let $f: A \rightarrow B$ be a bijection then, a function $g: B \rightarrow A$ which associates each element $b \in B$ to a different element $a \in A$ such that $f(a) = b$ is called the inverse of f .

$$f(a) = b \leftrightarrow g(b) = a$$

Composition of Functions

Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be two functions then, a function $g \circ f: A \rightarrow C$ is defined by

$$(g \circ f)(x) = g(f(x)), \text{ for all } x \in A$$

In-Text Questions (ITQs)

1. Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$, given by $f(x) = 2x$, is one-one and onto.
2. If $f : \mathbb{Q} \rightarrow \mathbb{Q}$ is given by $f(x) = x^2$, then find $f^{-1}(16)$.
3. If $f : A \rightarrow B$ and $|A| = 5$ and $|B| = 3$ then find total number of functions.

In-Text Answers (ITAs)

1. Let $a, b \in \mathbb{R}$ such that $f(a) = f(b)$ then,

$$f(a) = f(b)$$

$$\Rightarrow 2a = 2b$$

$$\Rightarrow a = b$$

Therefore, $f: \mathbb{R} \rightarrow \mathbb{R}$ is one-one.

For onto:

Let p be any real number in $\mathbb{R}$ (co-domain). $f(x) = p$

$$\Rightarrow 2x = p$$

$$\Rightarrow x = p/2$$

$$p/2 \in \mathbb{R} \text{ for } p \in \mathbb{R} \text{ such that } f(p/2) = 2(p/2) = p$$

For each $p \in \mathbb{R}$ (codomain) there exists $x = p/2 \in \mathbb{R}$ (domain) such that $f(x) = y$

For each element in codomain has its pre-image in domain.

So, $f: \mathbb{R} \rightarrow \mathbb{R}$ is onto.

Since $f: \mathbb{R} \rightarrow \mathbb{R}$ is both one-one and onto.

$f : \mathbb{R} \rightarrow \mathbb{R}$ is one-one correspondent (bijective function).

2. Let $f^{-1}(16) = x$

$$f(x) = 16$$

$$\Rightarrow x^2 = 16$$

$$\Rightarrow x = \pm 4$$

$$\text{Thus, } f^{-1}(16) = \{-4, 4\}$$

$$3. \text{ Total number of functions} = 35 = 243$$

4.1.3 Functions in Discrete Mathematics

Let X and Y be two sets with m and n elements and a function is defined as $f: X \rightarrow Y$ then,

- Total number of functions = nm
- Total number of one-one function = nP_m
- Total number of onto functions = $nm - nC_1(n-1)^m + nC_2(n-2)^m - \dots + (-1)^{n-1}nC_{n-1}1^m$ if $m \geq n$.

For the composition of functions f and g be two functions:

- $f \circ g \neq g \circ f$
- If f and g are both one-to-one functions, then $f \circ g$ is also one-to-one.
- If f and g are both onto functions, then $f \circ g$ is also onto.
- If f and $f \circ g$ both are one-to-one functions, then g is also one-to-one.
- If f and $f \circ g$ are both onto functions, then it is not necessary that g is also onto.

- $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$
- $f^{-1} \circ f = f^{-1}(f(a)) = f^{-1}(b) = a$
- $f \circ f^{-1} = f(f^{-1}(b)) = f(a) = b$

4.1.4 Injective vs Surjective

A function is injective if every element of the codomain is the image of at most one element from the domain.

A function is surjective provided every element of the codomain is the image of at least one element from the domain.

4.1.5 Conclusion

Functions form the backbone of discrete structures by providing formal rules for input-output mappings. Understanding their types and behaviours is crucial in logic,

computer science, and mathematics. Whether finite or infinite in domain and codomain, analysing functions helps in recognising patterns, solving problems, and building foundational knowledge for more advanced topics such as algorithms and set theory.

Summary

This unit explored the fundamental concept of functions in discrete mathematics. A function assigns a unique output to each input, defined clearly by its domain and codomain. The unit also classified different types of functions—injective, surjective, bijective, and their combinations—along with detailed examples. Graphical and matrix-based representations were introduced for better visualisation, especially with finite sets. Finally, the concepts of inverse and composite functions were explained, providing essential tools for analyzing relationships between sets.

Self-Assessment Questions

1. What distinguishes a bijective function from an injective or surjective function?
2. How is the composition of two functions $(f \circ g)(x)$ computed?
3. Explain why a function with more elements in its codomain than in its range is considered an "into" function.

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. Which functions are injective (i.e., one-to-one)?
 - a. $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(n) = 3n$.
 - b. $g: \{1, 2, 3\} \rightarrow \{a, b, c\}$ defined by $g = \begin{pmatrix} 1 & 2 & 3 \\ c & a & a \end{pmatrix}$.
 - c. $h: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ defined as follows:
2. Consider the function $g: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $g(n) = n^2 + 1$. Find $g(1)$ and $g(\{1\})$. Then find $g^{-1}(1)$, $g^{-1}(2)$, and $g^{-1}(3)$.

References/Further Readings

- Epp, S. S. (2010). Discrete Mathematics with Applications (4th ed.). Cengage Learning.
- Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.
- Grimaldi, R. P. (2004). Discrete and Combinatorial Mathematics: An Applied Introduction (5th ed.). Pearson.
- Kolman, B., Busby, R. C., & Ross, S. (2008). Discrete Mathematical Structures (6th ed.). Pearson Education.

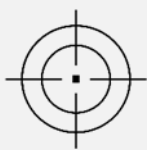
Module Five

Sequences and Summations

Unit 1 — Sequences

INTRODUCTION

Sequences and summations are fundamental concepts of mathematical analysis, and it has practical applications in science, engineering, and finance.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Define and describe sequences and distinguish between finite and infinite sequences.
- Identify and generate arithmetic and geometric progressions.
- Derive rules and general formulas from given sequences.
- Explain and solve basic recurrence relations with given initial conditions.

5.1.1 Definition

A sequence is a function from a subset of the set of integers (typically the set $\{0, 1, 2, \dots\}$ or the set $\{1, 2, 3, \dots\}$) to a set S . We use the notation a_n to denote the image of the integer n . We call a_n a term of the sequence.

A sequence is a function from a subset of the integers to a set S .

- A discrete structure used to represent an ordered list
- a_n denotes the sequence $a_0, a_1, a_2, \dots$
- Sometimes the sequence will start at a_1
- a_n is a term of the sequence

A **sequence** is a function from a subset of $\mathbb{Z}$ (usually $\{0, 1, 2, 3, \dots\}$ or $\{1, 2, 3, \dots\}$) to a set S . We use a_n to refer to the image of the integer n . We call a_n a **term** of the sequence. The sequence itself is denoted $\{a_n\}$.

It is a set of numbers in a definite order according to some definite rule (or rules). Each number of the set is called a term of the sequence and its length is the number of terms in it. We can write the sequence as: $\{a_n\}_{n=1}^{\infty}$ or a_n . A finite sequence is generally described by $a_1, a_2, a_3, \dots, a_n$, and an infinite sequence is described by $a_1, a_2, a_3, \dots$ to infinity. A sequence $\{a_n\}$ has the limit L and we write:

$$\lim_{n \rightarrow \infty} a_n = L \text{ or } a_n \rightarrow L \text{ as } n \rightarrow \infty$$

• Example

- 1, 2, 3, 5, 8 is a sequence with five terms
- 1, 3, 9, 27, ..., 3^n , ... is an infinite sequence
- What is the function which generates the terms of the sequence 5, 7, 9, 11, ... ?
 - If $a_0 = 5$
 - $a_n = 5 + 2n$
 - If $a_1 = 5$
 - $a_n = 3 + 2n$

Example:

- 2, 4, 6, 8, ..., 20 is a finite sequence obtained by adding 2 to the previous number.
- 10, 6, 2, -2, is an infinite sequence obtained by subtracting 4 from the previous number

5.1.2 Progression

If a formula can describe the terms of a sequence, then the sequence is called a progression.

1, 1, 2, 3, 5, 8, 13, ..., is a progression called the Fibonacci sequence in which each term is the sum of the previous two numbers.

5.1.3 Arithmetic Progression

Definition: An arithmetic progression is a sequence of the form

$$a, a+d, a+2d, \dots, a+nd$$

where a is the initial term and d is common difference, such that both belong to $\mathbb{R}$ (real numbers).

An arithmetic progression is a sequence of the form $a, a + d, a + 2d, \dots$ where the initial term a and the common difference d are real numbers.

Definition: A *arithmetic progression* is a sequence of the form:

$$a, a + d, a + 2d, \dots, a + nd, \dots \quad a_n = a + nd$$

where *initial term* a and *common difference* d are real numbers.

Example:

- $-1, 3, 7, 11, \dots$
- What is the initial term a and common difference d ?
 - $a = -1$
 - $d = 4$
 - $a_n = a + dn$
 - $a_n = -1 + 4n$

5.1.4 Geometric Progression

Definition: A geometric progression is a sequence of the form:

$$a, ar, ar^2, \dots, ar^k,$$

where a is the initial term, and r is the common ratio. Both a and r belong to $\mathbb{R}$

- A geometric progression is a sequence of the form

$$a, ar, ar^2, ar^3, \dots$$

where the initial term a and the common ratio r are real numbers ($\mathbb{R}$).

Example

- $a_n = \left(\frac{1}{2}\right)^n$ for $n = 0, 1, 2, 3, \dots$
 members: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$

Example

- $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$
- What is the initial term a and common ratio r ?
 - $a = 1$
 - $r = \frac{1}{2}$
 - $a_n = ar^n$
 - $a_n = \left(\frac{1}{2}\right)^n$

In-Text Questions (ITQs)

1. Let $a = -1$ and $d = 4$. Then:

What is $\{s_n\}$

2. Let $a = 1$ and $r = -1$. Then:

What is $\{b_n\}$

In-Text Answers (ITAs)

- 1.

$$\{s_n\} = \{s_0, s_1, s_2, s_3, s_4, \dots\} = \{-1, 3, 7, 11, 15, \dots\}$$

- 2.

$$\{b_n\} = \{b_0, b_1, b_2, b_3, b_4, \dots\} = \{1, -1, 1, -1, 1, \dots\}$$

5.1.5 Generating a rule from a sequence

Given a sequence, finding a rule for generating the sequence is not always straightforward

Example:

Question: Assume the sequence: $1, 3, 5, 7, 9, \dots$, What is the formula for the sequence?

Answer: Each term is obtained by adding 2 to the previous term. $1, 1+2=3, 3+2=5, 5+2=7$, What type of progression does this suggest?

It suggests an arithmetic progression: $a+nd$ with $a=1$ and $d=2$ $a_n=1+2n$

Given a sequence, finding a rule for generating the sequence is not always straightforward

Example:

Question: Assume the sequence: $1, 1/3, 1/9, 1/27, \dots$ What is the sequence?

Answer: The denominators are powers of 3. i.e. $1, 1/3=1/3, (1/3)/3=1/(3*3)=1/9, (1/9)/3=1/27$

This suggests a geometric progression: ar^k

with $a=1$ and $r=1/3$ i.e. $(1/3)^n$

5.1.6 Recurrence Relation

Definition: A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a non-negative integer.

- A sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.
- The initial conditions for a sequence specify the terms that precede the first term where the recurrence relation takes effect.
- A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence.

Example

Let $\{a_n\}$ be a sequence that satisfies the recurrence

relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, 4, \dots$ and suppose that $a_0 = 2$.
What are a_1, a_2 and a_3 ?

[Here $a_0 = 2$ is the initial condition.]

Solution: We see from the recurrence relation that:

$$a_1 = a_0 + 3 = 2 + 3 = 5$$

$$a_2 = 5 + 3 = 8$$

$$a_3 = 8 + 3 = 11$$

Example

Let $\{a_n\}$ be a sequence that satisfies the recurrence

relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$ and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3 ?

[Here the initial conditions are $a_0 = 3$ and $a_1 = 5$.] **Solution:** We see from the recurrence relation that:

$$a_2 = a_1 - a_0 = 5 - 3 = 2$$

$$a_3 = a_2 - a_1 = 2 - 5 = -3$$

- Geometric progression: $a_n = ar^n$
 - Recurrence relation: $a_n = ra_{n-1}$, $a_0 = a$ (a_0 is the initial condition)
- Arithmetic progression: $a_n = a + dn$
 - Recurrence relation: $a_n = a_{n-1} + d$, $a_0 = a$
- Fibonacci Sequence: $\{f_n\} = 0, 1, 1, 2, 3, 5, 8, 13, \dots$
 - $f_n = f_{n-1} + f_{n-2}$, $f_0 = 0$, $f_1 = 1$
- Factorials: $1, 1, 2, 6, 24, 120, 720, 5040, \dots$
 - $n! = n(n-1)(n-2) \cdots 2 \cdot 1$
 - $a_n = na_{n-1}$, $a_0 = 1$

Technique for Solving Recurrence Relations

Finding a formula for the n th term of the sequence generated by a recurrence relation is called solving the recurrence relation.

• Iteration

- Forward: work forward from the initial term until a pattern emerges. Then guess the form of the solution.
- Backward: work backward from a_n toward a_0 until a pattern emerges. Then guess the form of the solution.
- Example: $a_n = a_{n-1} + 3$, $a_0 = 2$
 - Forward
 - $a_1 = 2 + 3$
 - $a_2 = (2 + 3) + 3 = 2 + 2 \cdot 3$
 - $a_3 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$
 - ...
 - $a_n = 2 + 3n$
 - Backward
 - $a_n = (a_{n-2} + 3) + 3 = a_{n-2} + 2 \cdot 3$
 - $a_n = (a_{n-3} + 3) + 2 \cdot 3 = a_{n-3} + 3 \cdot 3$
 - $a_n = (a_{n-4} + 3) + 3 \cdot 3 = a_{n-4} + 4 \cdot 3$
 - ...
 - $a_n = a_{n-n} + n \cdot 3 = a_0 + 3n = 2 + 3n$

5.1.7 Conclusion

Sequences and summations form the foundation for more advanced mathematical concepts used in analysis, algorithms, and modeling. Understanding how to define, generate, and analyze sequences, including arithmetic and geometric progressions, equips learners with tools to solve real-world problems involving patterns and predictions.

Summary

This unit focused on the concept of sequences, which are ordered lists of numbers defined by a rule. It covered types of sequences including arithmetic progression (AP), geometric progression (GP), and Fibonacci sequence. Learners explored how to generate sequences using formulas and how to derive these formulas from patterns. The unit also introduced recurrence relations, which define terms in a sequence based on previous terms, and the basics of solving them.

Self-Assessment Questions

1. What distinguishes an arithmetic progression from a geometric progression? Give one example of each.
2. Identify the rule governing the sequence: 4, 8, 16, 32,... and write its general term.
3. What are the initial conditions in a recurrence relation, and why are they important?

Unit Activities

1. What distinguishes an arithmetic progression from a geometric progression? Give one example of each.
2. Identify the rule governing the sequence: 4, 8, 16, 32,... and write its general term.
3. What are the initial conditions in a recurrence relation, and why are they important?

Tutor-Marked Assignment

1. Consider the arithmetic sequence 7, 12, 17, 22,...
 - (a) Determine the 10th term of the sequence.
 - (b) Write a general formula for the n th term
2. A sequence is defined by the recurrence relation:
$$a_1 = 2, a_2 = 5, a_n = a_{n-1} + a_{n-2} \text{ for } n \geq 3$$
 - (a) Find the first 6 terms of the sequence.
 - (b) Explain the pattern or rule behind this sequence

References/Further Readings

- Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.
- Epp, S. S. (2011). Discrete Mathematics with Applications (4th ed.). Cengage Learning.
- Enderton, H. B. (2001). A Mathematical Introduction to Logic (2nd ed.). Academic Press

Module Six

Proof Techniques

Unit 1 — Proof Techniques

INTRODUCTION

A proof is a method for establishing the truth of a statement.

A mathematical proof is a verification for establishing the truth of a proposition by a chain of logical deductions from a set of axioms

Anyone who doesn't believe mathematics is creative has not tried to write proofs. Finding a way to convince the world that a particular statement is necessarily true is a mighty undertaking and can often be quite challenging. There is no guaranteed path to success in the search for proofs.

Writing proofs is a bit of an art. Like any art, to be truly great at it, you need some sort of inspiration, as well as some foundational technique. Just as musicians can learn proper fingering, and painters can learn the proper way to hold a brush, we can look at the proper way to construct arguments. A good place to start might be to study a classic.

Now that we have a solid understanding of propositional and predicate logic, it is time to actually put them to use. In this section we will examine a few proof methods strategies to formally prove something. Then, we will see lots of examples of putting them to use.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Define and describe sequences and distinguish between finite and infinite sequences.
- Identify and generate arithmetic and geometric progressions.
- Derive rules and general formulas from given sequences.

6.1.1 Some terminology

In mathematics, we have several different categories of “provable” things.

A **theorem** is a formal statement that can be shown to be true. It establishes a truth from which other theorems can be proved, or algorithms constructed.

Many famous mathematical theorems, e.g., the Pythagorean theorem, Fermat’s last

Pythagorean theorem: Let a , b be the lengths of the two sides of a right triangle, and let c be the hypotenuse.

Then, $a^2 + b^2 = c^2$

Fermat’s Last Theorem: For any integer n greater than 2, the equation $a^n + b^n = c^n$ has no solutions for non-zero a , b , c .

A **lemma** is a formal statement that can be shown to be true, and is used as part of the proof of another theorem. A lemma is a ‘helping theorem’ or a result that is needed to prove a theorem

A **proposition** is a formal statement that can be shown to be true, but the importance of this fact is less than that of a theorem. Less important theorems are sometimes called propositions.

A **corollary** is a special case of a theorem, lemma, or proposition. It follows directly from a theorem and is not typically proved explicitly. Corollaries are stated to show an “important result” or “important implication” of a theorem. A corollary is a result that follows directly from a theorem.

A **conjecture** is a formal statement that is likely to be true or is being proposed to be true, but has not yet been proved to be true. Once proven to be true, it may become a formal theorem or lemma

6.1.2 Types of Proofs / General Strategies for Proving Theorems

Suppose we wish to prove an implication $p \rightarrow q$. Here are some strategies we have available to try

- **Trivial Proof:**

If we know q is true, then $p \rightarrow q$ is true regardless of the truth value of p

Example

If it is raining, then $1=1$. (Proposition)

Proof:

This implication is trivially true since $1=1$ is known to be true. The truth value of the premise does not matter.

Example -

If there are 1000 employees in an organisation, then $3^2 = 9$.

Explanation -

Let P: There are 1000 employees in an organisation, & Q: $3^2 = 9$.

We know that Q is always true, and in the truth table we can see that whenever Q is true, $P \Rightarrow Q$ is true, whatever the true value of P is.

Another obvious proof for implication $p \rightarrow q$ relies on p being false. Recall, "falsity implies anything". If we know p to be false, then the implication $p \rightarrow q$ is always true. A vacuous proof is thus a simple statement that shows p to be false.

▪ **Vacuous Proof:**

If p is a conjunction of other hypotheses and we know one or more of these hypotheses is false, then p is false and so $p \rightarrow q$ is vacuously true regardless of the truth value of q.

Example

If the door is both open and closed, $2+2=5$. (Proposition)

Proof:

This implication is trivially true since the premise is always false (by the law of complementation). The truth value of the conclusion does not matter.

- If P is a conjunction (example : $P = A \wedge B \wedge C$) of other hypotheses and we know one or more of these hypotheses is false, then P is false and so $P \rightarrow Q$ is vacuously true regardless of the truth value of Q.

Example -

If $5 \neq 100$, then $3 \neq 6$.

Explanation -

Let P : $5 \neq 100$, & Q : $3 \neq 6$.

We know that P is always false and in the truth table we can see that whenever P is False, $P \Rightarrow Q$ is true, whatever the truth value of Q is.

Direct Proof:

Assume p, and then use the rules of inference, axioms, definitions, and logical

The simplest (from a logic perspective) style of proof is a direct proof. Often all that is required to prove something is a systematic explanation of what everything means. Direct proofs are especially useful when proving implications. The general format to prove $P \rightarrow Q$ is this: Assume P . Explain, explain, ..., explain. Therefore Q .

Often we want to prove universal statements, perhaps of the form $\forall x (P(x) \rightarrow Q(x))$. Again, we will want to assume $P(x)$ is true and deduce $Q(x)$. But what about the x ? We want this to work for all x . We accomplish this by fixing x to be an arbitrary element (of the sort we are interested in).

The first strategy you should try when attempting to prove any assertion is to give a direct proof. That is, assume the hypotheses that are given and try to argue directly that the conclusion follows. This is often the best approach when the hypotheses can be translated into algebraic expressions (equations or inequalities) that can be manipulated to give other algebraic expressions, which are useful in verifying the conclusion.

Example

Prove the statement: For all integers m and n , if m and n are odd integers, then $m + n$ is an even integer.

Proof.

Assume m and n are arbitrary odd integers. Then m and n can be written in the form

$$m = 2a + 1 \text{ and } n = 2b + 1,$$

where a and b are also integers. Then

$$m + n = (2a + 1) + (2b + 1) \text{ (substitution)}$$

$$= 2a + 2b + 2 \text{ (associative and commutative laws of addition)}$$

$$= 2(a + b + 1) \text{ (distributive law)}$$

Since $m+n$ is twice another integer, namely, $a+b+1$, $m+n$ is an even integer.

Example

Prove: For all integers n , if n is even, then n^2 is even.

Solution

The format of the proof will be this: Let n be an arbitrary integer. Assume that n is even. Explain explain explain. Therefore n^2 is even.

To fill in the details, we will basically just explain what it means for n to be even, and then see what that means for n^2 . Here is a complete proof.

Proof

Let n be an arbitrary integer. Suppose n is even. Then $n=2k$ for some integer k . Now $n^2=(2k)^2=4k^2=2(2k^2)$. Since $2k^2$ is an integer, n^2 is even.

- **Indirect Proof or Proof by Contradiction:**

Assume p and $\neg q$ and derive a contradiction $r \wedge \neg r$.

- **Proof by Contrapositive: (Special case of Proof by Contradiction.)**

Give a direct proof of $\neg q \rightarrow \neg p$. Assume $\neg q$ and then use the rules of inference, axioms, definitions, and logical equivalences to prove $\neg p$. (Can be thought of as a proof by contradiction in which you assume p and $\neg q$ and arrive at the contradiction $p \wedge \neg p$.)

- **Proof by Cases: If the hypothesis p can be separated into cases $p_1 \vee p_2 \vee \dots \vee p_k$, prove each of the propositions,**

$p_1 \rightarrow q, p_2 \rightarrow q, \dots, p_k \rightarrow q$, separately.

(You may use different methods of proof for different cases.)

Proof by Cases

In some cases, it is very difficult to prove a theorem by applying the same argument in all cases

For example, we might need to consider different arguments for negative and non-negative integers

Proof by cases allows us to apply different arguments in different cases and combine the results

Specifically, suppose we want to prove statement p , and we know that we have either q or r . If we can show $q \rightarrow p$ and $r \rightarrow p$, then we can conclude p

In general, there may be more than two cases to consider

....

In a theorem, it is possible that the premise itself is a compound statement. The compound statement may be explicit, like "Let $a=1$ or $a=-1$ ", or implicit, like "let n be an integer". In this latter statement, there are different

kinds of cases you could apply. You could consider n to be odd or even. You may consider n to be zero, positive, or negative.

There are two steps in a proof by cases:

- Extract from the theorem statement the possible cases; and
- Prove each case independently.

The validity of a proof by cases comes from the following logical equivalence:

$$(p_1 \vee p_2 \vee \cdots \vee p_n) \rightarrow q \equiv (p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \cdots \wedge (p_n \rightarrow q)$$

$$(p_1 \vee p_2 \vee \cdots \vee p_n) \rightarrow q \equiv (p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \cdots \wedge (p_n \rightarrow q)$$

Each of the “or” conditions in the premise is a separate case that we must prove.

....

If the hypothesis p can be separated into cases $p_1 \vee p_2 \vee \cdots \vee p_k$, prove each of the propositions, $p_1 \rightarrow q, p_2 \rightarrow q, \dots, p_k \rightarrow q$, separately.

$p_1 \rightarrow q, p_2 \rightarrow q, \dots, p_k \rightarrow q$, separately.

(You may use different methods of proof for different cases.)

Example

Proposition

There is a natural number n such that $n^2 + 3n + 2$ is prime

Proof

False!

Negation. $\forall$ natural number n , $n^2 + 3n + 2$ is composite.

We prove the negation in two cases:

1. n is even
2. n is odd

Prove that n is even

$\Rightarrow n^2 + 3n + 2$ is composite. n is even

$\Rightarrow n^2$ is even and $3n$ is even (even $\times$ integer = even)

$\Rightarrow n^2 + 3n + 2$ is even (even + even = even)

$\Rightarrow n^2 + 3n + 2$ is composite (2 is a factor)

Prove that n is odd

$\Rightarrow n^2 + 3n + 2$ is composite. n is odd
 $\Rightarrow n^2$ is odd and $3n$ is odd (odd $\times$ odd = odd)
 $\Rightarrow n^2 + 3n$ is even (odd + odd = even)
 $\Rightarrow n^2 + 3n + 2$ is even (even + even = even)
 $\Rightarrow n^2 + 3n + 2$ is composite (2 is a factor)

Example

Prove: For any integer n , the number $(n^3 - n)$ is even.

Solution

It is hard to know where to start this, because we don't know much of anything about n . We might be able to prove that $n^3 - n$ is even if we knew that n was even. In fact, we could probably prove that $n^3 - n$ was even if n was odd. But since n must either be even or odd, this will be enough. Here's the proof.

Proof

We consider two cases: if n is even or if n is odd.

Case 1: n is even. Then $n = 2k$ for some integer k . This gives

$$n^3 - n = 8k^3 - 2k$$

$$= 2(4k^2 - k),$$

and since $4k^2 - k$ is an integer, this says that $n^3 - n$ is even.

Case 2: n is odd. Then $n = 2k + 1$ for some integer k . This gives

$$n^3 - n = (2k + 1)^3 - (2k + 1)$$

$$= 8k^3 + 6k^2 + 6k + 1 - 2k - 1$$

$$= 2(4k^3 + 3k^2 + 2k),$$

and since $4k^3 + 3k^2 + 2k$ is an integer, we see that $n^3 - n$ is even again.

Since $n^3 - n$ is even in both exhaustive cases, we see that $n^3 - n$ is indeed always even.

Definition (even and odd integers): An integer m is even if it satisfies the formula $m = 2k$ for some integer k . An integer n is odd if it satisfies the formula $n = 2k + 1$ for some integer k .

Definition (rational number): A number n is rational if there exists integers p and q such that $n=p/q$ and $q \neq 0$.

In the following two examples, we prove the statement through a direct proof. By assuming p is true, and then showing q must also be true, we have directly proven $p \rightarrow q$.

Example

Proposition

Sum of an even integer and an odd integer is odd.

(ie Even + odd = odd)

Proof

Suppose a is even and b is odd.

$$\begin{aligned}
 \text{Then } a + b &= (2m) + b && \text{(defn. of even, } a = 2m \text{ for integer } m) \\
 &= (2m) + (2n + 1) && \text{(defn. of odd, } b = 2n + 1 \text{ for integer } n) \\
 &= 2(m + n) + 1 && \text{(taking 2 as common factor)} \\
 &= 2p + 1 && (p = m + n \text{ and addition is closed on integers)} \\
 &= \text{odd} && \text{(defn. of odd)}
 \end{aligned}$$

Example

Proposition

The square of an odd integer is odd.

(n is odd $\Rightarrow n^2$ is odd)

Proove:

If n is odd, then n^2 is odd. n is odd

$$\begin{aligned}
 \Rightarrow n &= (2k + 1) && \text{(defn. of odd, } k \text{ is an integer)} \\
 \Rightarrow n^2 &= (2k + 1)^2 && \text{(squaring on both sides)} \\
 \Rightarrow n^2 &= 4k^2 + 4k + 1 && \text{(expanding the binomial)} \\
 \Rightarrow n^2 &= 2(2k^2 + 2k) + 1 && \text{(factoring 2 from first two terms)} \\
 \Rightarrow n^2 &= 2j + 1 && \text{(let } j = 2k^2 + 2k \text{) } (j \text{ is an integer as mult. and add. are closed on integers)}
 \end{aligned}$$

$\Rightarrow n^2$ is odd (defn. of odd)

In-Text Questions (ITQs)

1. Write a pattern that proves that every odd integer is equal to the difference between the squares of two integers
2. (Odd = difference of squares)

Prove: For all integers a , b , and c , if $a|b$ and $b|c$ then $a|c$. Here $x|y$, read “ x divides y ” means that y is a multiple of x (so x will divide into y without remainder).

3. For all integers p and q , if p and q are odd integers, then $p + q$ is an even integer.

Let P denotes: p and q are odd integers

Q : $p + q$ is an even integer

To Prove: $P \Rightarrow Q$

In-Text Answers (ITAs)

1. Write a formal statement.

$\forall$ integer k , $\exists$ integers m , n such that

$$(2k + 1) = m^2 - n^2.$$

Try out a few examples.

$$1 = 1^2 - 0^2$$

$$-1 = 0^2 - (-1)^2$$

$$3 = 2^2 - 1^2$$

$$-3 = (-1)^2 - (-2)^2$$

$$5 = 3^2 - 2^2$$

$$-5 = (-2)^2 - (-3)^2$$

$$7 = 4^2 - 3^2$$

$$-7 = (-3)^2 - (-4)^2$$

Find a pattern.

$$(k + 1)^2 - k^2 = (k^2 + 2k + 1) - k^2 = 2k + 1 = \text{odd}$$

2. Even before we know what the divides symbol means, we can set up a direct proof for this statement. It will go something like this: Let a , b , and c be arbitrary integers. Assume that $a|b$ and $b|c$. Dot dot dot. Therefore $a|c$.

How do we connect the dots? We say what our hypothesis ($a|b$ and $b|c$) really means and why this gives us what the conclusion ($a|c$) really means. Another way to say that $a|b$ is to say that $b=ka$ for some integer k (that is,

that b is a multiple of a). What are we going for? That $c=la$, for some integer l (because we want c to be a multiple of a).

Here is the complete proof.

Proof

Let a , b , and c be integers. Assume that $a|b$ and $b|c$. In other words, b is a multiple of a and c is a multiple of b . So there are integers k and j such that $b=ka$ and $c=jb$. Combining these (through substitution) we get that $c=jka$. But jk is an integer, so this says that c is a multiple of a . Therefore $a|c$.

3. As p & q are odd integers, they can be represented as :

Assume : $p = 2m + 1$ and $q = 2n + 1$, where m & n are also some integers.

Then : $p + q =$

$$= (2m + 1) + (2n + 1) \quad \text{(Substitution Law)}$$

$$= 2m + 2n + 2 \quad \text{(associative and commutative law for addition)}$$

$$= 2(m + n + 1) \quad \text{(distributive law)}$$

$$= \text{Number divisible by 2 \& hence an even number.}$$

6.1.3 Conclusion

Proof techniques form the foundation of mathematical reasoning. They help validate statements logically and rigorously. Mastery of different proof methods equips learners to approach mathematical problems confidently, ensuring their solutions are not just correct but also justified through accepted logical frameworks.

Summary

This unit introduced the fundamental concept of mathematical proofs and their importance in establishing truth in mathematics. It explored different proof strategies trivial, vacuous, direct, indirect, contrapositive, and proof by cases demonstrating how they are employed to confirm statements using logical reasoning and algebraic manipulation. Key terms such as theorems, lemmas, corollaries, propositions, and conjectures were also defined to contextualize the hierarchy of provable mathematical statements.

Self-Assessment Questions

1. Prove that the square of an integer is odd if and only if the integer itself is odd. Prove that there is no largest integer.
(n is odd $\Leftrightarrow n^2$ is odd)
2. Prove the statement: For all integers m and n , if the product of m and n is even, then m is even or n is even.
(We prove the contrapositive of the statement: If m and n are both odd integers, then mn is odd.)
3. If $x^3 - 7x^2 + x - 7 = 0$, then $x \neq 10$. Prove
4. Let x and y be real numbers. If $5a + 25b = 156$, then a or b is not an integer. Prove.

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. Let x and y be real numbers. If $5a + 25b = 156$, then a or b is not an integer. Prove
2. Prove that there is no largest integer

References/Further Readings

- Epp, S. S. (2011). Discrete Mathematics with Applications (4th ed.). Brooks/Cole.
- Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.
- Gries, D., & Schneider, F. B. (1993). A Logical Approach to Discrete Math. Springer.
- Hammack, R. (2013). Book of Proof (2nd ed.). Virginia Commonwealth University.
- Susanna S. Epp (2003). The Role of Logic in Teaching Proof. The American Mathematical Monthly, 110(10), 886–899 /Cole Cengage Learning

Module Seven

Mathematical Induction

Unit 1 — Mathematical Induction

INTRODUCTION

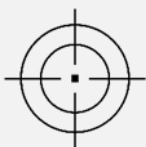
Mathematical induction is a proof technique, not unlike direct proof or proof by contradiction or combinatorial proof. In other words, induction is a style of argument we use to convince ourselves and others that a mathematical statement is always true. Many mathematical statements can be proved by simply explaining what they mean.

Suppose we have an infinite ladder, and we know two things:

1. We can reach the first rung of the ladder
2. If we reach a particular rung, then we can also reach the next rung

From these two facts, can we conclude we can reach every step of the infinite ladder?

Answer is yes, and mathematical induction allows us to make arguments like this.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Explain the concept of mathematical induction and its significance in proof construction.
- Apply the three-step process (base case, inductive hypothesis, inductive step) to prove mathematical statements.
- Use mathematical induction to prove summation formulas and divisibility properties.
- Relate real-world analogies such as ladders and dominoes to understand inductive reasoning.

7.1.1 What is Mathematical Induction

Mathematical Induction is one of the fundamental methods of writing proofs and it is used to prove a given statement about any well-organized set. Generally, it is used for proving results or establishing statements that are formulated in terms of n , where n is a natural number.

Suppose $P(n)$ is a statement for n natural number then it can be proved using the Principle of Mathematical Induction, Firstly we will prove for $P(1)$ then let $P(k)$ be true then prove for $P(k+1)$. If $P(k+1)$ holds then we say that $P(n)$ is true by the principle of mathematical induction.

We can compare mathematical induction to falling dominoes. When a domino falls, it knocks down the next domino in succession. The first domino knocks down the second one, the second one knocks down the third, and so on. In the end, all of the dominoes will be bowled over. But there are some conditions to be fulfilled:

- The base step is that the starting domino must fall to set the knocking process in action.
- The distance between dominoes must be equal for any two adjacent dominoes. Otherwise, a certain domino may fall down without bowling over the next. Then the sequence of reactions will stop. Maintaining the equal inter-domino distance ensures that $P(k) \Rightarrow P(k + 1)$ for each integer $k \geq a$. This is the inductive step.

7.1.2 Principle of Mathematical Induction Statement / Mathematics

Any statement $P(n)$ which is for " n " natural number can be proved using the Principle of Mathematical Induction by following the below steps,

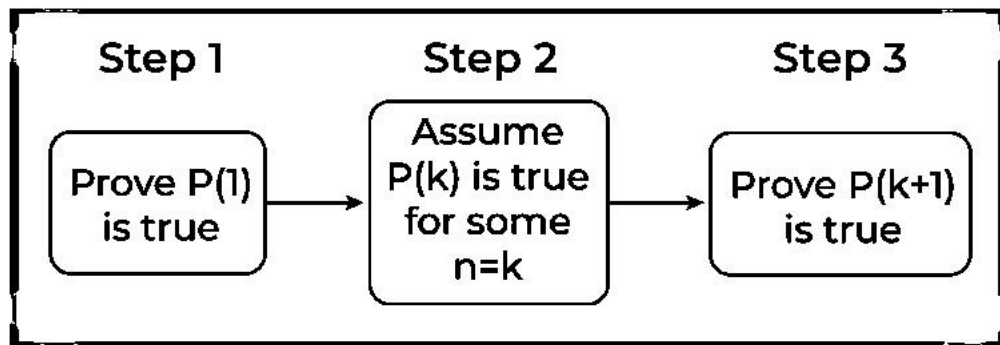
Step 1: Verify if the statement is true for trivial cases ($n = 1$) i.e. check if $P(1)$ is true.

Step 2: Assume that the statement is true for $n = k$ for some $k \geq 1$ i.e. $P(k)$ is true.

Step 3: If the truth of $P(k)$ implies the truth of $P(k + 1)$, then the statement $P(n)$ is true for all $n \geq 1$.

The image added below contains all the steps of Mathematical Induction

Principle of Mathematical Induction



The first statement is the fact and if it is not possible for all $P(n)$ to hold true at $n = 1$ then these statements are true for some other values of n say $n = 2, n = 3$, and others.

If the statement is true for $P(k)$ then if $P(k+1)$ is proven to be true then we say that $P(n)$ is true for all n belonging to Natural Numbers (N)

Example:

For any positive integer number n , prove that $n^3 + 2n$ is always divisible by 3

Let $P(n)$: $n^3 + 2n$ is divisible by 3 be the given statement.

Step 1: Basic Step

Firstly we prove that $P(1)$ is true. Let $n = 1$ in $n^3 + 2n$

$$= 1^3 + 2(1)$$

$$= 3$$

As 3 is divisible by 3. Hence, $P(1)$ is true.

Step 2: Assumption Step

Let us assume that $P(k)$ is true

Then, $k^3 + 2k$ is divisible by 3

Thus, we can assume $k^3 + 2k = 3n$, (where n is any positive integer)....(i)

Step 3: Induction Steps

Now we have to prove that algebraic expression $(k + 1)^3 + 2(k + 1)$ is divisible by 3

$$= (k + 1)^3 + 2(k + 1)$$

$$= k^3 + 3k^2 + 3k + 2k + 3$$

$$= (k^3 + 2k) + (3k^2 + 3k + 3)$$

from eq(i)

$$\begin{aligned}
 &= 3n+3(k^2+ k + 1) \\
 &= 3(n + k^2 + k + 1)
 \end{aligned}$$

As it is a multiple of 3 we can say that it is divisible by 3.

Thus, $P(k+1)$ is true i.e. $(k + 1)^3 + 2(k + 1)$ is be divisible by 3. Now by the Principle of Mathematical Induction, we can say that, $P(n): n^3 + 2n$ is divisible by 3 is true.

Example

Prove for each natural number $n \geq 1$ that $1+2+3+\dots+n=n(n+1)/2$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}.$$

First, let's think inductively about this equation. In fact, we know this is true for other reasons (reverse and add comes to mind). But why might induction be applicable? The left-hand side adds up the numbers from 1 to n . If we know how to do that, adding just one more term $(n+1)$ would not be that hard. For example, if $n=100$, suppose we know that the sum of the first 100 numbers is 5050 (so $1+2+3+\dots+100=5050$, which is true). Now to find the sum of the first 101 numbers, it makes more sense to just add 101 to 5050, instead of computing the entire sum again. We would have $1+2+3+\dots+100+101=5050+101=5151$. In fact, it would always be easy to add just one more term. This is why we should use induction.

Now the formal proof:

Proof

Let $P(n)$ be the statement

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}.$$

We will show that $P(n)$ is true for all natural numbers $n \geq 1$.

Base case: $P(1)$ is the statement

$$1 = \frac{1(1+1)}{2}$$

which is clearly true.

Inductive case: Let $k \geq 1$ be a natural number. Assume (for induction) that $P(k)$ is true. That means

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

We will prove that $P(k+1)$ is true as well. That is, we must prove that

$$1 + 2 + 3 + \dots + k + (k+1) = \frac{(k+1)(k+2)}{2}$$

To prove this equation, start by adding $k+1$ to both sides of the inductive hypothesis:

$$1 + 2 + 3 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1).$$

Now, simplifying the right side we get:

$$\begin{aligned} \frac{k(k+1)}{2} + k+1 &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+2)(k+1)}{2}. \end{aligned}$$

Thus $P(k+1)$ is true, so by the principle of mathematical induction $P(n)$ is true for all natural numbers $n \geq 1$

In-Text Questions (ITQs)

1. $1+3+5+\dots+(2n-1) = n^2$ for $n=1, 2, \dots$ Prove
2. Prove that for all $n \in \mathbb{N}$, $6^n - 1$ is a multiple of 5.

In-Text Answers (ITAs)

1. **Step 1** – For $n=1, 1=1^2$ Hence, step 1 is satisfied.

Step 2 – Let us assume the statement is true for $n=k$.

Hence, $1+3+5+\dots+(2k-1) = k^2$ is true (It is an assumption)

We have to prove that

$1+3+5+\dots+(2(k+1)-1) = (k+1)^2$ also holds

$$1+3+5+\dots+(2(k+1)-1) = 1+3+5+\dots+(2k-1) + (2k+1)$$

$$= 1+3+5+\dots+(2k-1) + (2k+1)$$

$$= 1+3+5+\dots+(2k-1) + (2k+1)$$

$$= 1+3+5+\dots+(2k-1) + (2k+1)$$

$$= k^2 + (2k+1) = (k+1)^2$$

$$=(k+1)2=(k+1)2$$

So, $1+3+5+\dots+(2(k+1)-1)=(k+1)2$ $1+3+5+\dots+(2(k+1)-1)=(k+1)2$ hold which satisfies the step 2.

Hence, $1+3+5+\dots+(2n-1)=n^2$ is proved.

2. Again, start by understanding the dynamics of the problem. What does increasing n do? Let's try with a few examples. If $n=1$, then yes, $6^1-1=5$ is a multiple of 5. What does incrementing n to 2 look like? We get $6^2-1=35$, which again is a multiple of 5. Next, $n=3$: but instead of just finding 6^3-1 , what did the increase in n do? We will still subtract 1, but now we are multiplying by another 6 first. Viewed another way, we are multiplying a number which is one more than a multiple of 5 by 6 (because 6^2-1 is a multiple of 5, so 6^2 is one more than a multiple of 5). What do numbers which are one more than a multiple of 5 look like? They must have last digit 1 or 6. What happens when you multiply such a number by 6? Depends on the number, but in any case, the last digit of the new number must be a 6. And then if you subtract 1, you get last digit 5, so a multiple of 5.

The point is, every time we multiply by just one more six, we still get a number with last digit 6, so subtracting 1 gives us a multiple of 5. Now the formal proof:

Proof

Let $P(n)$ be the statement, " 6^n-1 is a multiple of 5." We will prove that $P(n)$ is true for all $n \in \mathbb{N}$.

Base case: $P(0)$ is true: $6^0-1=0$ which is a multiple of 5.

Inductive case: Let k be an arbitrary natural number. Assume, for induction, that $P(k)$ is true. That is, 6^k-1 is a multiple of 5. Then $6^k-1=5j$ for some integer j . This means that $6^k=5j+1$. Multiply both sides by 6:

$$6^{k+1} = 6(5j + 1) = 30j + 6.$$

But we want to know about $6^{k+1}-1$, so subtract 1 from both sides:

$$6^{k+1} - 1 = 30j + 5.$$

Of course, $30j+5=5(6j+1)$, so is a multiple of 5.

Therefore

$$6^{k+1} - 1$$

is a multiple of 5, or in other words, $P(k+1)$ is true. Thus, by the principle of mathematical induction $P(n)$ is true for all $n \in \mathbb{N}$.

Prove the following statement by induction:

$$\forall n \in \mathbb{Z}^+. \sum_{i=1}^n i = \frac{(n)(n+1)}{2}$$

Base case: $n = 1$. In this case, $\sum_{i=1}^1 i = 1$ and $\frac{(1)(1+1)}{2} = 1$; thus, the base case holds.

Inductive step: By the inductive hypothesis, we assume $P(k)$:

$$\sum_{i=1}^k i = \frac{(k)(k+1)}{2}$$

Now, we want to show $P(k+1)$:

$$\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$$

First, observe:

$$\sum_{i=1}^{k+1} i = \sum_{i=1}^k i + (k+1)$$

By the inductive hypothesis, $\sum_{i=1}^k i = \frac{(k)(k+1)}{2}$; thus:

$$\sum_{i=1}^{k+1} i = \frac{(k)(k+1)}{2} + (k+1)$$

Rewrite left hand side as:

$$\sum_{i=1}^{k+1} i = \frac{k^2 + 3k + 2}{2} = \frac{(k+1)(k+2)}{2}$$

Since we proved both base case and inductive step, property holds.

7.1.3 Conclusion

Mathematical induction plays a crucial role in discrete mathematics, particularly in proving generalizations over natural numbers. It combines logic with structured reasoning, ensuring the correctness of mathematical assertions from the ground up. Mastering this technique enhances problem-solving and analytical skills essential for advanced mathematical thinking.

Summary

Mathematical induction is a powerful method for proving statements about natural numbers. It involves verifying the truth of a base case and then showing that if the statement holds for an arbitrary value k , it must also hold for $k+1$. This recursive structure ensures the statement is true for all natural numbers. This unit provided foundational understanding and multiple examples, from algebraic sums to divisibility proofs, to build learners' confidence in applying this technique.

Self-Assessment Questions

1. What are the three major steps in the principle of mathematical induction?
2. How does the inductive step ensure the validity of a statement for all $n \in \mathbb{N}$?
3. Why is the base case essential in a mathematical induction proof?

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. Prove that $n^2 < 2^n$ for all integers $n \geq 5$.
2. **Show that $3|(n^3 - n)$ whenever n is a positive integer.**

References/Further Readings

- Epp, S. S. (2011). Discrete Mathematics with Applications (4th ed.). Brooks/Cole.
- Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.
- Hammack, R. (2013). Book of Proof (2nd ed.). Virginia Commonwealth University.
- Kolman, B., Busby, R. C., & Ross, S. (2008). Discrete Mathematical Structures (6th ed.). Pearson Education.
- Leithold, L. (2002). The Calculus 7 (7th ed.). Brooks/Cole. (For induction in summation and series).

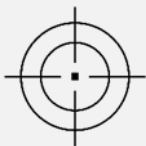
Module Eight

Inclusion-Exclusion

Unit 1 — Inclusion-exclusion

INTRODUCTION

Inclusion and Exclusion is an important concept in Set Theory that helps in calculating the cardinality of the union of multiple sets; this is True even when those sets overlap. The basic idea is to include elements in the union but exclude those that have been counted multiple times with the overlapping sets. It's a bit complex at first, but we'll break it down and understand how it works.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Explain the Principle of Inclusion and Exclusion (PIE) in counting problems.
- Apply PIE to calculate the cardinality of the union of two or more overlapping sets.
- Use Venn diagrams to visualize set intersections and unions.
- Clearly state and apply the Pigeonhole Principle to solve counting and distribution problems.
- Distinguish between injective and surjective mappings using set theory concepts.
- Analyse real-life scenarios using the Pigeonhole Principle to deduce logical conclusions.

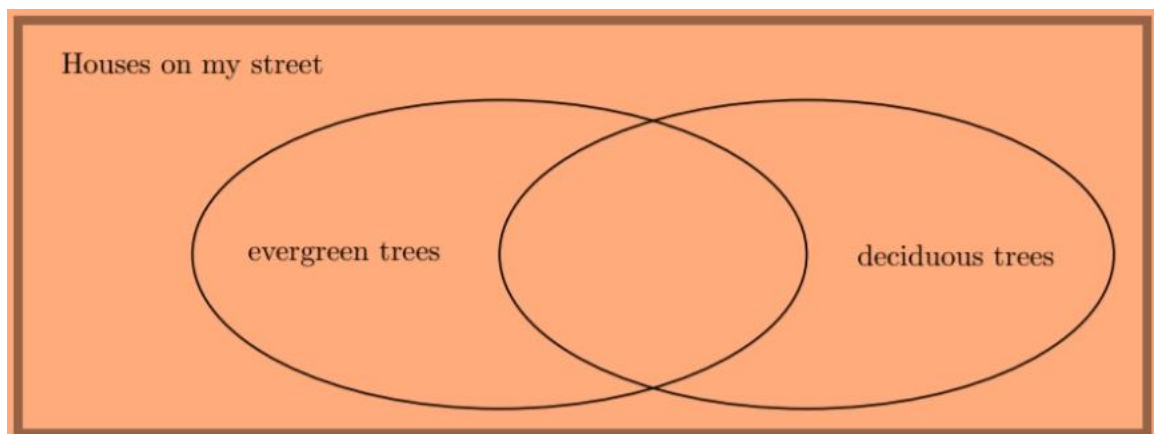
8.1.1 What is Mathematical Induction?

It is a smart way to count the number of items in overlapping groups without accidentally counting some items more than once. It works by first adding the number of items in each group, then subtracting the number of items in any two groups that overlap, and so on.

For example, if you have three groups of students studying different subjects, PIE helps figure out how many students study at least one subject by adjusting for those who study more than one, so nobody is counted twice.

The Principle of Inclusion and Exclusion (PIE) is useful for finding the size of the union of two or more sets. When these sets overlap, adding their sizes together this may lead to double-counting the elements. PIE corrects this by subtracting the sizes of the intersections. But if multiple intersections overlap, further adjustments are needed.

We could draw a very basic Venn diagram showing the kinds of trees that are growing at the various houses on my street:



Looking at the Venn diagram can help us figure out the values of some of the pieces from knowing the values of others. Suppose we know how many houses have deciduous trees, and how many houses have evergreen trees. Naively, you might think that adding these together would give us the total number of houses with trees. However, by looking at the Venn diagram, we see that if we simply add the values together, then any houses that have both kinds of trees have been counted twice (once as a house with a deciduous tree, and again as a house with an evergreen tree). So in order to work out the number of houses that have trees, we can add the number that have deciduous trees to the number that have evergreen trees and

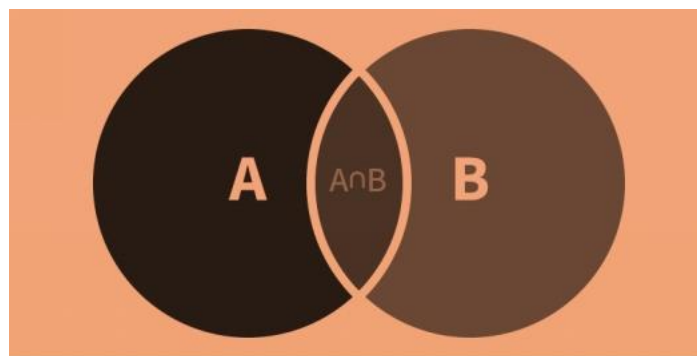
then subtract the number that have both kinds of trees. This is the idea of “inclusion-exclusion.”

Specifically, if two sets A and B are disjoint, then $|A \cup B| = |A| + |B|$. However, if A and B are not disjoint, then $|A| + |B|$ counts the elements of $A \cap B$ twice (both as elements of A and as elements of B). Subtracting this overcount yields the correct answer:

Proposition: Inclusion for 2 sets

For any finite sets A and B , we have

$$|A \cup B| = |A| + |B| - |A \cap B|.$$



Proof

Let $A_0 = A \setminus B$ and $B_0 = B \setminus A$, so

- A is the disjoint union of A_0 and $A \cap B$,
- B is the disjoint union of B_0 and $A \cap B$, and
- $A \cup B = (A_0 \cup (A \cap B)) \cup (B_0 \cup (A \cap B))$ is the disjoint union of A_0 , B_0 , and $A \cap B$.

Then

$$\begin{aligned} |A| + |B| &= (|A_0| + |A \cap B|) + (|B_0| + |A \cap B|) \\ &= (|A_0| + |B_0| + |A \cap B|) + |A \cap B| \\ &= |A \cup B| + |A \cap B| \end{aligned}$$

Example

Let $A = \{p, r, o, n, g\}$ and $B = \{h, o, r, n, s\}$. Then,

$$|A| = 5, |B| = 5, \text{ and } |A \cap B| = |\{r, o, n\}| = 3$$

so Inclusion-Exclusion tells us that

$$|A \cup B| = |A| + |B| - |A \cap B| = 5 + 5 - 3 = 7$$

This is correct, since

$$|A \cup B| = |\{p, r, o, n, g, h, s\}| = 7$$

Proposition: Inclusion-Exclusion for three sets:

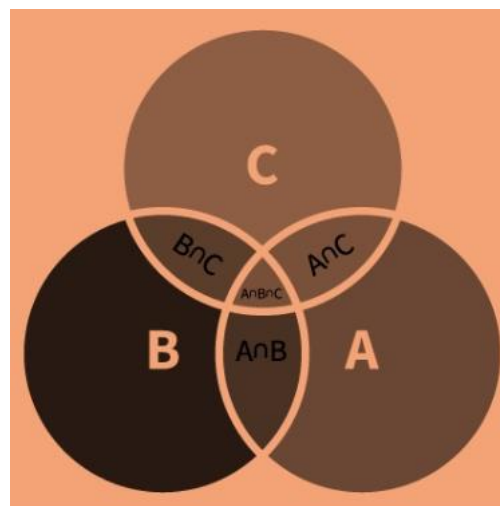
For three sets A, B, C the Principle of Inclusion and Exclusion (PIE) formula to find the size of the union is:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Where,

- $|A|, |B|, |C|$ are number of elements in the sets A, B and C respectively.
- $|A \cap B|, |A \cap C|, |B \cap C|$ are the numbers of elements in the pairwise intersections of the sets.
- $|A \cap B \cap C|$ is the number of elements that are in all three sets.

3.0.3 Venn Diagram for Three Sets



- **First Step (Inclusion):** Add the sizes of the individual sets $|A|$, $|B|$, and $|C|$. This counts all the elements in A, B, and C, but it also counts the elements in the intersections more than once.
- **Second Step (Exclusion):** Subtract the pairwise intersections $|A \cap B|$, $|B \cap C|$, and $|C \cap A|$ because these elements were counted twice in the first step.

- **Third Step (Inclusion of Intersection of All Three Sets):** Add back the intersection of all three sets $|A \cap B \cap C|$, as it was subtracted three times (once for each pairwise intersection) in the previous step.
- Thus, the number of elements in the union of these three sets, $|A \cup B \cup C|$, is given by:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$

8.1.2 Approach

Inclusion-Exclusion Principle is a combinatorial counting technique that allows us to count the number of elements in the union of multiple sets. The principle states that the size of the union of two or more sets is equal to the sum of their sizes minus the size of their intersection, plus the size of the intersection of their pairwise intersections, and so on.

Here's the step-by-step approach in C++ to implement the Inclusion-Exclusion Principle:

- Define the sets that need to be combined.
- Compute the size of each set.
- Compute the size of each intersection of two sets.
- Compute the size of each intersection of three sets.
- Continue computing the size of each intersection of four, five, and so on sets until you reach the final intersection.
- Sum the sizes of all sets.
- Subtract the size of all pairwise intersections.
- Add the size of all three-way intersections.
- Continue adding and subtracting the intersections of increasing sizes until you reach the final intersection.
- Return the final count.

Here is an example of implementing the Inclusion-Exclusion Principle in C++ & Python3 to find the number of positive integers less than 100 that are divisible by either 2, 3, or 5:

For C++

```
#include <iostream>

using namespace std;
```

```
int main() {  
  
    int n = 100;  
  
    int count = 0;  
  
    // Count the number of integers divisible by 2  
    for(int i = 2; i < n; i += 2) {  
        count++;  
    }  
  
    // Count the number of integers divisible by 3  
    for(int i = 3; i < n; i += 3) {  
        count++;  
    }  
  
    // Count the number of integers divisible by 5  
    for(int i = 5; i < n; i += 5) {  
        count++;  
    }  
  
    // Count the number of integers divisible by both 2 and 3  
    for(int i = 6; i < n; i += 6) {  
        count--;  
    }  
  
    // Count the number of integers divisible by both 2 and 5  
    for(int i = 10; i < n; i += 10) {
```

```

        count--;
    }

    // Count the number of integers divisible by both 3 and 5
    for(int i = 15; i < n; i += 15) {
        count--;
    }

    // Count the number of integers divisible by 2, 3, and 5
    for(int i = 30; i < n; i += 30) {
        count++;
    }

    // Print the final count

    cout << "The number of positive integers less than " << n <<
    " that are divisible by either 2, 3, or 5 is " << count << "." << endl;

    return 0;
}

```

For Python3

```

n = 100

count = 0

# Count the number of integers divisible by 2
for i in range(2, n, 2):

```

```
count += 1
```

```
# Count the number of integers divisible by 3
```

```
for i in range(3, n, 3):
```

```
    count += 1
```

```
# Count the number of integers divisible by 5
```

```
for i in range(5, n, 5):
```

```
    count += 1
```

```
# Count the number of integers divisible by both 2 and 3
```

```
for i in range(6, n, 6):
```

```
    count -= 1
```

```
# Count the number of integers divisible by both 2 and 5
```

```
for i in range(10, n, 10):
```

```
    count -= 1
```

```
# Count the number of integers divisible by both 3 and 5
```

```
for i in range(15, n, 15):
```

```
    count -= 1
```

```
# Count the number of integers divisible by 2, 3, and 5
```

```
for i in range(30, n, 30):
```

```
    count += 1
```

```
# Print the final count
```

```
print(f"The number of positive integers less than {n} that are divisible by  
either 2, 3, or 5 is {count}.")
```

8.1.3 Theorem: Inclusion-Exclusion

Let $A_1, \dots, A_n$ be finite sets. Then

$$|A_1 \cup \dots \cup A_n| = \left(\sum_{i=1}^n |A_i| \right) - \left(\sum_{1 \leq i < j \leq n} |A_i \cap A_j| \right) + \dots + ((-1)^{n+1} |A_1 \cap \dots \cap A_n|)$$

Of course, we can figure out the value of any one of the terms in the inclusion-exclusion formula, if we know the values of all of the other

In-Text Questions (ITQs)

1. In the Faculty of Arts and Science, the voting method used is "approval;" that is, regardless of the number of positions available, each voter can mark as many boxes as they wish on their ballot.
2. Imagine that Prof. Li, Prof. Cheng, and Prof. Osborn were all nominated for two computer science positions on the department's search committee. Barb Hodgson notes the following facts when counting the ballots:
 - a. Prof. Cheng received 1818 votes; Prof. Osborn received 1515 votes, and Prof. Li received 10 votes.
 - b. Only one ballot had all three boxes marked.
 - c. Five of the ballots were marked for both Prof. Osborn and Prof. Li.
 - d. Ten of the ballots were marked for Prof. Cheng and Prof. Osborn.
 - e. Six of the ballots were marked for Prof. Cheng and Prof. Li.

How many members of the department voted in the election?

3. For any positive integer number n , prove that $n^3 + 2n$ is always divisible by 3
 Let $P(n)$: $n^3 + 2n$ is divisible by 3 be the given statement.

In-Text Answers (ITAs)

- Again, we begin by establishing some notation. Let C be the set of ballots that were marked for Prof. Cheng; let O be the set of ballots that were marked for Prof. Osborn; and let L be the set of ballots that were marked for Prof. Li. Then what we want is $|C \cup O \cup L|$: the number of ballots that were marked for at least one of the three candidates; this is the same as the number of people who voted.

Inclusion-Exclusion tells us that

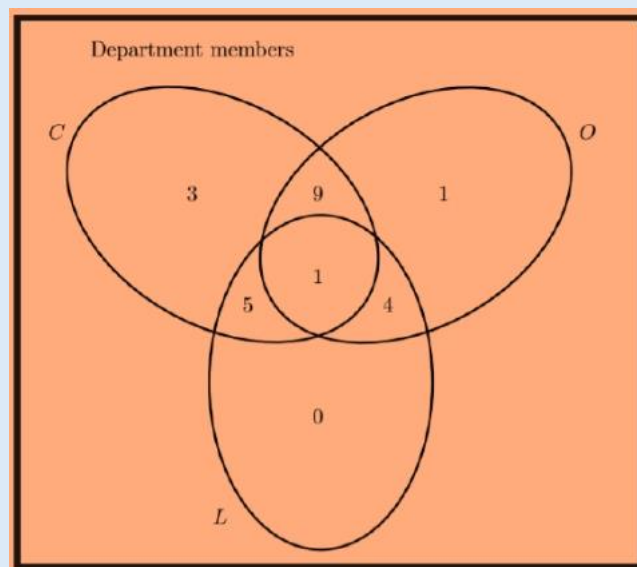
$$|C \cup O \cup L| = |C| + |O| + |L| - |C \cap O| - |C \cap L| - |O \cap L| + |C \cap O \cap L|$$

We have been given all of the values on the right-hand side of this equation, so we see that

$$|C \cup O \cup L| = 18 + 15 + 10 - 10 - 6 - 5 + 1 = 23$$

There were 23 department members who voted in the election.

In fact, the information we have been given is enough for us to fill in the values in every piece of the Venn diagram.



The 9 people who voted for Prof. Cheng and Prof. Osborn but not Prof. Li is determined from the fact that 10 people voted for Professors Cheng and Osborn, and only one of those voted for all three professors. Similarly, the 4 people who voted for Prof. Li and Prof. Osborn but not Prof. Cheng is determined from the fact that 5 people voted for Professors Li and Osborn, and only one of those voted for all three professors; also, the 55 people who voted for Prof. Cheng and Prof. Li but not Prof. Osborn is determined from the fact that 6 people voted for Professors Cheng and Li, and only one of those voted for all three professors.

From the above deductions, we see that of the 18 votes Prof. Cheng received, one ballot was marked for all 3 candidates; 9 were marked for Professors Cheng and Osborn (but not Li); and 5 were marked for Professors Cheng and Li (but not Osborn). The remaining 3 votes must have been for Prof. Cheng alone, allowing us to fill in that spot. Similarly, of the 15 votes Prof. Osborn received, one ballot was marked for all 3 candidates; 9 were marked for Professors Cheng and Osborn (but not Li); and 4 were marked for Professors Osborn and Li (but not Cheng). The remaining vote must have been for Prof. Osborn alone, allowing us to fill in that spot. Finally, all of Prof. Li's 10 votes are accounted for between the 5 who voted for Professors Cheng and Li (but not Osborn), the 4 who voted for Professors Li and Osborn (but not Cheng) and the one who voted for all three, so we put a 0 into the final spot.

2. **Step 1: Basic Step**

Firstly we prove that $P(1)$ is true. Let $n = 1$ in $n^3 + 2n$

$$= 1^3 + 2(1)$$

$$= 3$$

As 3 is divisible by 3. Hence, $P(1)$ is true

Step 2: Assumption Step

Let us assume that $P(k)$ is true

Then, $k^3 + 2k$ is divisible by 3

Thus, we can assume $k^3 + 2k = 3n$, (where n is any positive integer)....(i)

3. **Step 3: Induction Steps**

Now we have to prove that algebraic expression $(k + 1)^3 + 2(k + 1)$ is divisible by 3

$$= (k + 1)^3 + 2(k + 1)$$

$$= k^3 + 3k^2 + 3k + 2k + 2$$

$$= (k^3 + 2k) + (3k^2 + 3k + 2)$$

from eq(i)

$$= 3n + 3(k^2 + k + 1)$$

$$= 3(n + k^2 + k + 1)$$

As it is a multiple of 3 we can say that it is divisible by 3.

Thus, $P(k+1)$ is true i.e. $(k + 1)^3 + 2(k + 1)$ is be divisible by 3. Now by the Principle of Mathematical Induction, we can say that, $P(n): n^3 + 2n$ is divisible by 3 is true.

8.1.4 Pigeonhole Principles

The Pigeonhole Principle is a fundamental concept in combinatorics and mathematics that states if more items are put into fewer containers than the number of items, at least one container must contain more than one item. This seemingly simple principle has profound implications and applications in various fields, including mathematics, computer science, and engineering.

What is Mathematical Induction

The Pigeonhole Principle can be formally stated as follows:

If n items are distributed among m containers and $n > m$, then at least one container must contain more than one item.

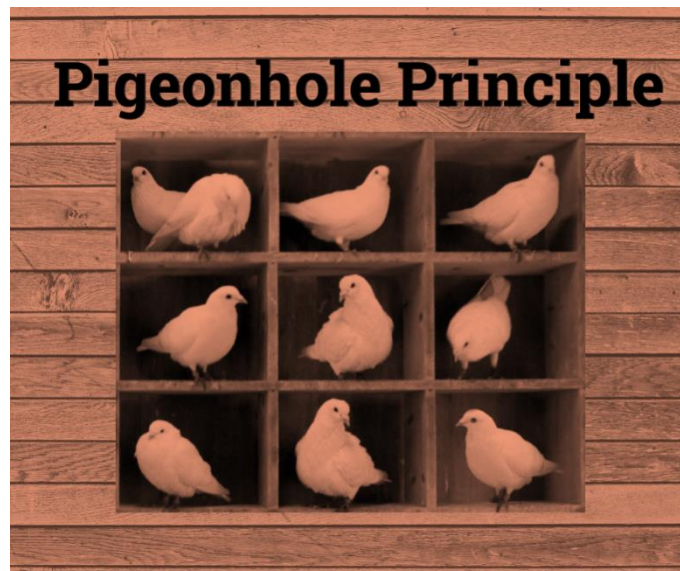
The pigeonhole principle is useful in counting methods. In order to apply the principle, one has to decide which objects will play the role of pigeon and which objects will play the role of pigeonholes.

Imagine a situation where there are 20 pigeons trying to enter into 19 pigeonholes. Since there are more pigeons than pigeonholes, at least one of the pigeonholes must contain more than one pigeon. This straightforward idea lays the foundation for many mathematical proofs.

If k objects are placed into n pigeonholes, and $k > n$, then at least one pigeonhole must contain more than one object.

If 10 pigeons are located in 9 pigeonholes, then there is a pigeonhole with more than one pigeon.

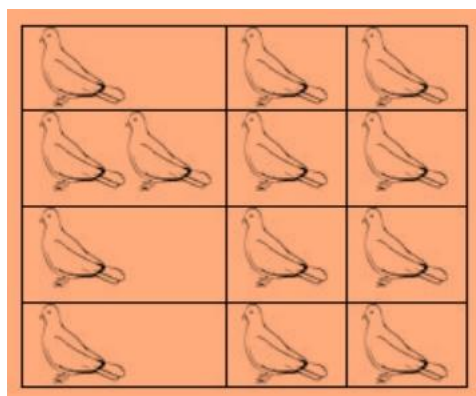
- If a set A has more elements than a set B , then a map $F : A \rightarrow B$ cannot be injective (one-to-one). That is, some elements a_1, a_2 from A will have the same image.
- If a set A has fewer elements than a set B , then a map $F : A \rightarrow B$ cannot be surjective (onto). That is, some element b from B has empty preimage



Examples

- In a class of 13 students, at least two must be born in the same month. Here, the 13 students are “pigeons” and the 12 months are “pigeonholes”.
- If 102 students took an exam with maximal score 100 points, then at least two students will have the same score. The students are “pigeons”, the numbers of points are “pigeonholes”.
- In a letter with 30 words at least two words begin with the same letter. The words are “pigeons” and the 26 letters are “pigeonholes”.
- Among 100 integers $a_1, \dots, a_{100}$ one can find two $a_i, a_j, i \neq j$, whose difference is divisible by 97. Integers $a_1, \dots, a_{100}$ are “pigeons”, residues mod 97 are “pigeonholes”.
- A drawer contains 10 pairs of socks of different colors and you pick some randomly. What minimum number guarantees a pair of one color? The 10 colors are “pigeonholes”, so we need to pick 11 to guarantee.

Pigeonhole Principle : If k is a positive integer and $k + 1$ objects are placed into k boxes, then at least one of the boxes will contain two or more objects



Example

For any choice of six digits in the set $S = \{1, 2, \dots, 9\}$ one can find two chosen digits giving in sum 10.

Solution: Pigeons here are digits and pigeonholes are 5 subsets $\{1, 9\}$, $\{2, 8\}$, $\{3, 7\}$, $\{4, 6\}$, $\{5\}$. Among six chosen digits two will be in the same subset, and thus, give in sum 10.

Example

If there are $n > 1$ people who can shake hands with one another, then there is always two persons who will shake hands with the same number of people.

Solution: Each person shakes hands from 0 to $n - 1$ people, totally n possibilities. But 0 means that someone shakes hands to nobody, while $n - 1$ means shaking hands to everybody. So, both 0 and $n - 1$ cannot happen at the same time and this leaves n people to $n - 1$ possibilities

In-Text Questions (ITQs)

1. Find the minimum number of students in a class such that three of them are born in the same month?
2. If $(Kn+1)$ pigeons are kept in n pigeon holes where K is a positive integer, what is the average no. of pigeons per pigeon hole?

In-Text Answers (ITAs)

1. According to the given condition,

$$K+1=3$$

$$K=2$$

$$M=kn+1=2 \times 12+1=25.$$
2. Average number of pigeons per hole $= (Kn+1)/n = K + 1/n$
 Therefore there will be at least one pigeonhole which will contain at least $(K+1)$ pigeons i.e., $\text{ceil}[K + 1/n]$ and remaining will contain at most K i.e., $\text{floor}[k+1/n]$ pigeons. i.e., the minimum number of pigeons required to ensure that at least one pigeon hole contains $(K+1)$ pigeons is $(Kn+1)$

8.1.5 Conclusion

The inclusion-exclusion principle is a practical tool for accurately computing the total number of unique elements across intersecting sets. It is foundational in solving complex combinatorial and probabilistic problems in both mathematical theory and computer algorithms. Through real-world analogies, worked examples, and coding demonstrations, learners gain an intuitive and computational grasp of PIE. Understanding the Pigeonhole Principle allows learners to approach combinatorial problems more strategically. Whether in classroom exercises or in abstract mathematical proofs, the principle reveals how structure and limitation lead to predictable outcomes—an essential reasoning tool in discrete mathematics and computer science.

Summary

The Principle of Inclusion and Exclusion (PIE) is a key concept in set theory and combinatorics used to avoid overcounting elements in overlapping sets. When combining the sizes of multiple sets, direct addition can lead to counting shared elements more than once. PIE addresses this by subtracting the sizes of intersections between sets and appropriately adjusting for overlaps involving more than two sets. This unit introduced PIE for two and three sets with theoretical explanations, Venn diagram representations, worked examples, and computational implementation in Python and C++. The Pigeonhole Principle asserts that if more items are distributed across fewer containers, then at least one container must hold more than one item. This simple yet powerful concept finds applications in set theory, number theory, computer science, and real-world problem-solving. Through multiple illustrative examples, learners gain practical understanding and confidence in applying the principle both in academic and everyday contexts.

Self-Assessment Questions

1. What problem does the Principle of Inclusion and Exclusion solve?
2. State the PIE formula for two and three sets.
3. Describe how Venn diagrams help in understanding inclusion and exclusion.
4. What is the basic condition that must be met for the Pigeonhole Principle to apply?

5. In a room of 367 people, why must at least two people share the same birthday?
6. Explain how the generalized Pigeonhole Principle can be used to prove that in a group of 51 people, at least five were born in the same month

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. **Apply PIE** to determine how many numbers between 1 and 200 are divisible by 4, 5, or 6.
2. Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $B = \{3, 4, 5, 6\}$, and $C = \{5, 6, 7, 8\}$. Find $|A \cup B \cup C|$ using the inclusion-exclusion principle.
3. Write a Python or C++ program to compute how many integers between 1 and 500 are divisible by either 3, 7, or 11.
4. Show that from any three integers, one can always choose two so that $a^3b - ab^3$ is divisible by 10.
5. A drawer contains 15 black, 13 blue, and 12 white socks. What is the minimum number of socks you must pick to guarantee getting at least two socks of the same color? Justify your answer using the Pigeonhole Principle.

References/Further Readings

- Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.
- Epp, S. S. (2011). Discrete Mathematics with Applications (4th ed.). Brooks/Cole.
- Hammack, R. (2013). Book of Proof (2nd ed.). Virginia Commonwealth University.
- Kolman, B., Busby, R. C., & Ross, S. (2008). Discrete Mathematical Structures (6th ed.). Pearson Education.
- Tucker, A. (2012). Applied Combinatorics. John Wiley & Sons.
- Rosen, K. H. (2019). Discrete Mathematics and Its Applications (8th ed.). McGraw-Hill.
- Grimaldi, R. P. (2004). Discrete and Combinatorial Mathematics: An Applied Introduction. Pearson Education.
- Epp, S. S. (2010). Discrete Mathematics with Applications. Brooks/Cole Cengage Learning.

Module Nine

Permutations and Combinations (with and without repetitions)

Unit 1 — Permutations and Combinations

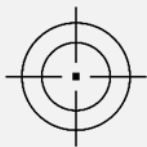
INTRODUCTION

A permutation is a (possible) rearrangement of objects. For example, there are 6 permutations of the letters a, b, c:

.abc, acb, bac, bca, cab, cba.

We know that we have them all listed above —there are 3 choices for which letter we put first, then 2 choices for which letter comes next, which leaves only 1 choice for the last letter. The multiplicative principle says we multiply $3 \cdot 2 \cdot 1$

.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Define and compute permutations and combinations using factorial notation.
- Distinguish between when to use permutations versus combinations.
- Apply permutation and combination formulas to real-world and theoretical problems.

9.1.1 Function

A piece of notation is helpful here: $n!$, read “ n factorial”, is the product of all positive integers less than or equal to n (for reasons of convenience, we also define $0!$ to be 1). So the number of permutation of 6 letters, as seen in the previous example is $6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$. This generalizes:

Permutations of n elements.

There are $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$ permutations of n (distinct) elements.

There are $n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$ permutations of n (distinct) elements.

Example (Counting Bijective Functions)

How many functions $f: \{1, 2, \dots, 8\} \rightarrow \{1, 2, \dots, 8\}$ are bijective?

Remember what it means for a function to be bijective: each element in the codomain must be the image of exactly one element of the domain. Using two-line notation, we could write one of these bijections as

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 1 & 5 & 8 & 7 & 6 & 2 & 4 \end{pmatrix}.$$

What we are really doing is just rearranging the elements of the codomain, so we are creating a permutation of 8 elements. In fact, “permutation” is another term used to describe bijective functions from a finite set to itself.

If you believe this, then you see the answer must be

$$8! = 8 \cdot 7 \cdot \cdots \cdot 1 = 40320.$$

You can see this directly as well: for each element of the domain, we must pick a distinct element of the codomain to map to. There are 8 choices for where to send 1, then 7 choices for where to send 2, and so on. We multiply using the multiplicative principle.

Example

How many 4 letter “words” can you make from the letters a through f, with no repeated letters?

This is just like the problem of permuting 4 letters, only now we have more choices for each letter. For the first letter, there are 6 choices. For each of those, there are 5

choices for the second letter. Then there are 4 choices for the third letter, and 3 choices for the last letter. The total number of words is

$$6 \cdot 5 \cdot 4 \cdot 3 = 360.$$

This is not $6!$ because we never multiplied by 2 and 1. We could start with $6!$ and then cancel the 2 and 1, and thus write

$$\frac{6!}{2!}$$

in general, we can ask how many permutations exist of k objects choosing those objects from a larger collection of n objects. (In the example above, $k=4$, and $n=6$.) We write this number $P(n,k)$ and sometimes call it a k -permutation of n elements. From the example above, we see that to compute $P(n,k)$ we must apply the multiplicative principle to k numbers, starting with n and counting backwards. For example

$$P(10,4) = 10 \cdot 9 \cdot 8 \cdot 7.$$

Notice again that $P(10,4)$ starts out looking like $10!$, but we stop after 7. We can formally account for this “stopping” by dividing away the part of the factorial we do not want:

Careful: The factorial in the denominator is not $4!$ but rather

***k*-permutations of n elements.**

$P(n, k)$ is the number of ***k*-permutations of n elements**, the number of ways to *arrange* k objects chosen from n distinct objects.

$$P(n, k) = \frac{n!}{(n-k)!} = n(n-1)(n-2) \cdots (n-(k-1)).$$

- Note that when $n = k$, we have $P(n, n) = \frac{n!}{(n-n)!} = n!$ (since we defined $0!$ to be 1). This makes sense—we already know $n!$ gives the number of permutations of all n objects.

9.1.2 Combination

It is the distinct sections of a shared number of components carried one by one, or some, or all at a time. For example, if there are two components A and B, then there is only one way to select two things, select both of them.

For example, let $n = 3$ (A, B, and C) and $r = 2$ (All combinations of size 2). Then there are 3C_2 such combinations, which is equal to 3. These three combinations are AB, AC, and BC.

Here, the **combination** of any two letters out of three letters A, B, and C is shown below, we notice that in combination the order in which A and B are taken is not important as AB and BA represent the same combination.

Note: In the same example, we have distinct points for permutation and combination. For, AB and BA are two distinct items i.e., two distinct permutation, but for selecting, AB and BA are the same i.e., same combination.

An r -combination of set S is the unordered selection of r elements from that set

Unlike permutations, order does not matter in combinations

Example: What are 2-combinations of the set {a, b, c}?

For this set, 6 2-permutations, but only 3 2-combinations

The number of r -combinations of a set with n elements is written $C(n, r)$

$C(n, r)$ is often also written as

$$\binom{n}{r}$$

read "n choose r" and is also called the binomial coefficient

Theorem:

$$C(n, r) = \binom{n}{r} = \frac{n!}{r! \cdot (n - r)!}$$

Proof

What is the relationship between $P(n, r)$ and $C(n, r)$?

Let's decompose $P(n, r)$ using product rule:

First choose r elements

Then, order these r elements

How many ways to choose r elements from n ?

How many ways to order r elements? | Thus, $P(n, r) = C(n, r) * r!$

Therefore

$$C(n, r) = \frac{P(n, r)}{r!} = \frac{n!}{(n - r)! \cdot r!}$$

Combination Formula:

Combination Formula is used to choose 'r' components out of a total number of 'n' components, and is given by:

$${}^nC_r = \frac{n!}{r!(n - r)!}$$

Example : Find the number of permutations and combinations of $n = 9$ and $r = 3$.

Solution:

Given, $n = 9, r = 3$

Using the formula given above:

For Permutation:

$$\begin{aligned} nPr &= (n!) / (n - r)! \\ \Rightarrow nPr &= (9!) / (9 - 3)! \\ \Rightarrow nPr &= 9! / 6! = (9 \times 8 \times 7 \times 6!) / 6! \\ \Rightarrow nPr &= 504 \end{aligned}$$

For Combination:

$$\begin{aligned} nCr &= n! / r! (n - r)! \\ \Rightarrow nCr &= 9! / 3! (9 - 3)! \\ \Rightarrow nCr &= 9! / 3! (6)! \\ \Rightarrow nCr &= 9 \times 8 \times 7 \times 6! / 3! (6)! \\ \Rightarrow nCr &= 84 \end{aligned}$$

Example: In how many ways can a committee consisting of 4 men and 2 women can be chosen from 6 men and 5 women?

Solution:

Choose 4 men out of 6 men = 6C_4 ways = 15 ways

Choose 2 women out of 5 women = 5C_2 ways = 10 ways

The committee can be chosen in ${}^6C_4 \times {}^5C_2 = 150$ ways.

Example:

In how many ways can 5 different books be arranged on a shelf?

Solution:

This is a permutation problem because the order of the books matters.

Using the permutation formula, we get:

$${}_5P_5 = 5! / (5 - 5)! = 5! / 0! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Therefore, there are 120 ways to arrange 5 different books on a shelf.

In-Text Questions (ITQs)

1. In how many ways can 5 children be arranged in a line such that
 - i. two particular children of them are always together
 - ii. two particular children of them are never together.
2. A committee of 5 members is to be formed from a group of 10 people. In how many ways can this be done?

In-Text Answers (ITAs)

1.
 - i. We consider the arrangements by taking 2 particular children together as one and hence the remaining 4 can be arranged in $4! = 24$ ways. Again, two particular children taken together can be arranged in two ways. Therefore, there are $24 \times 2 = 48$ total ways of arrangement.
 - ii. Among the $5! = 120$ permutations of 5 children, there are 48 in which two children are together. In the remaining $120 - 48 = 72$ permutations, two particular children are never together.
2. This is a combination problem because the order of the members doesn't matter.

Using the combination formula, we get:

$${}^{10}C_5 = \frac{10!}{(5! \times (10 - 5)!)} = \frac{10!}{(5! \times 5!)} \\ \Rightarrow {}^{10}C_5 = \frac{(10 \times 9 \times 8 \times 7 \times 6)}{(5 \times 4 \times 3 \times 2 \times 1)} = 252$$

Therefore, there are 252 ways to form a committee of 5 members from a group of 10 people.

9.1.3 Conclusion

Understanding permutations and combinations is crucial for solving complex counting problems in mathematics, computer science, data analysis, and everyday reasoning. Recognizing whether order matters or not determines which technique to use. With proper understanding and practice, students can model, simplify, and solve a wide variety of arrangement and selection problems effectively.

Summary

This unit introduced the concepts of permutations and combinations, foundational tools in counting and probability theory. Permutations consider the arrangement of items where order matters, while combinations focus on selections where order is irrelevant. The unit included several real-life scenarios and mathematical examples to illustrate the application of these formulas in decision-making and problem-solving.

Self-Assessment Questions

1. In how many ways 3 mathematics books, 4 history books, 3 chemistry books and 2 biology books can be arranged on a shelf so that all books of the same subjects are together.
2. In how many ways 3 mathematics books, 4 history books, 3 chemistry books and 2 biology books can be arranged on a shelf so that all books of the same subjects are together.

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)

2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. A student has to answer 10 questions, choosing atleast 4 from each of Parts A and B. If there are 6 questions in Part A and 7 in Part B, in how many ways can the student choose 10 questions?
2. Out of 6 consonants and 3 vowels, how many expressions of 2 consonants and 1 vowel can be created?

References/Further Readings

- Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.
- Epp, S. S. (2010). Discrete Mathematics with Applications (4th ed.). Cengage Learning.
- Grimaldi, R. P. (2004). Discrete and Combinatorial Mathematics: An Applied Introduction (5th ed.). Pearson.
- Kolman, B., Busby, R. C., & Ross, S. (2008). Discrete Mathematical Structures (6th ed.). Pearson.

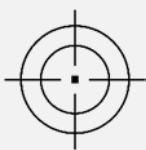
Module Ten

Binomial Theorem

Unit 1 — Binomial Theorem

INTRODUCTION

Binomial theorem is a fundamental principle in algebra that describes the algebraic expansion of powers of a binomial. According to this theorem, the expression $(a + b)^n$ where a and b are any numbers and n is a non-negative integer. It can be expanded into the sum of terms involving powers of a and b .



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Define and differentiate between permutations and combinations.
- Apply factorial notation in solving counting problems involving arrangements and selections.
- Compute the number of permutations and combinations for given values of n and r .
- Solve real-life problems involving selection and arrangement using permutation and combination formulas.

10.1.1 Binomial Theorem

Binomial theorem is used to find the expansion of two terms hence it is called the Binomial Theorem.

Binomial theorem is used to solve binomial expressions simply. This theorem was first used somewhere around 400 BC by Euclid, a famous Greek mathematician.

It gives an expression to calculate the expansion of algebraic expression $(a+b)^n$. The terms in the expansion of the following expression are exponent terms and the constant term associated with each term is called the coefficient of terms.

Statement: According to the binomial theorem, it is possible to expand any non-negative power of binomial $(x + y)$ into a sum of the form,

$$(x+y)^n = {}^nC_0 x^n y^0 + {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_{n-1} x^1 y^{n-1} + {}^nC_n x^0 y^n$$

where, $n \geq 0$ is an integer and each nC_k is a positive integer known as a binomial coefficient.

Note: When an exponent is zero, the corresponding power expression is 1. This multiplicative factor is often omitted from the term, therefore often the right-hand side is directly written as ${}^nC_0 x^n + \dots$. This formula is also referred to as the [binomial formula](#) or the binomial identity. Using summation notation, the binomial theorem can be given as,

$$(x+y)^n = \sum_{k=0}^n {}^nC_k x^{n-k} y^k = \sum_{k=0}^n {}^nC_k x^k y^{n-k}$$

Example:

Let us expand $(x+3)^5$ using the binomial theorem.

Here $y = 3$ and $n = 5$.

Substituting and expanding, we get:

$$\begin{aligned} (x+3)^5 &= {}^5C_0 x^5 3^0 + {}^5C_1 x^{5-1} 3^1 + {}^5C_2 x^{5-2} 3^2 + {}^5C_3 x^{5-3} 3^3 + {}^5C_4 x^{5-4} 3^4 + {}^5C_5 x^{5-5} 3^5 \\ &= x^5 + 5x^4 \cdot 3 + 10x^3 \cdot 9 + 10x^2 \cdot 27 + 5x \cdot 81 + 3^5 \\ &= x^5 + 15x^4 + 90x^3 + 270x^2 + 405x + 243 \end{aligned}$$

Binomial Expansion

The binomial theorem is also known as the binomial expansion which gives the formula for the expansion of the exponential power of a **binomial expression**. Binomial expansion of $(x + y)^n$ by using the binomial theorem is as follows,

$$(x+y)^n = {}^nC_0 x^n y^0 + {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_{n-1} x^1 y^{n-1} + {}^nC_n x^0 y^n$$

Binomial Theorem Formula

The binomial theorem formula is used in the expansion of any power of a binomial in the form of a series. The binomial theorem formula is

$$(a+b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r,$$

where n is a positive integer and a, b are real numbers, and $0 < r \leq n$.

This formula helps to expand the binomial expressions such as $(x + a)^{10}$, $(2x + 5)^3$, $(x - (1/x))^4$, and so on. The binomial theorem formula helps in the expansion of a binomial raised to a certain power. Let us understand the binomial theorem formula and its application in the following sections.

The binomial theorem states: If x and y are real numbers, then for all $n \in \mathbb{N}$,

$$(x+y)^n = {}^nC_0 x^n y^0 + {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_k x^{n-k} y^k + \dots + {}^nC_n x^0 y^n$$

$$\Rightarrow (x + y)^n = \sum_{k=0}^n {}^nC_k x^{n-k} y^k$$

$$\text{where, } {}^nC_r = n! / [r! (n - r)!]$$

Binomial Theorem Expansion Proof

Let $x, a, n \in \mathbb{N}$. Let us prove the binomial theorem formula through **the principle of mathematical induction**. It is enough to prove for $n = 1$, $n = 2$, for $n = k \geq 2$, and for $n = k + 1$.

It is obvious that $(x + y)^1 = x + y$ and

$$\begin{aligned} (x + y)^2 &= (x + y) (x + y) \\ &= x^2 + xy + xy + y^2 \text{ (using distributive property)} \\ &= x^2 + 2xy + y^2 \end{aligned}$$

Thus the result is true for $n = 1$ and $n = 2$. Let k be a positive integer. Let us prove the result is true for $k \geq 2$.

Assuming $(x + y)^n = \sum_{r=0}^n {}^nC_r x^{n-r} y^r$,

$$(x + y)^k = \sum_{r=0}^k {}^kC_r x^{k-r} y^r$$

$$\Rightarrow (x+y)^k = {}^kC_0 x^k y^0 + {}^kC_1 x^{k-1} y^1 + {}^kC_2 x^{k-2} y^2 + \dots + {}^kC_r x^{k-r} y^r + \dots + {}^kC_k x^0 y^k$$

$$\Rightarrow (x+y)^k = x^k + {}^kC_1 x^{k-1} y^1 + {}^kC_2 x^{k-2} y^2 + \dots + {}^kC_r x^{k-r} y^r + \dots + y^k$$

Thus the result is true for $n = k \geq 2$.

Now consider the expansion for $n = k + 1$.

$$(x + y)^{k+1} = (x + y) (x + y)^k$$

$$= (x + y) (x^k + {}^kC_1 x^{k-1} y^1 + {}^kC_2 x^{k-2} y^2 + \dots + {}^kC_r x^{k-r} y^r + \dots + y^k)$$

$$= x^{k+1} + (1 + {}^kC_1) x^k y + ({}^kC_1 + {}^kC_2) x^{k-1} y^2 + \dots + ({}^kC_{r-1} + {}^kC_r) x^{k-r+1} y^r + \dots + ({}^kC_{k-1} + 1) x y^k + y^{k+1}$$

$$= x^{k+1} + {}^{k+1}C_1 x^k y + {}^{k+1}C_2 x^{k-1} y^2 + \dots + {}^{k+1}C_r x^{k-r+1} y^r + \dots + {}^{k+1}C_k x y^k + y^{k+1} \text{ [Because } {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r]$$

Thus the result is true for $n = k+1$. By mathematical induction, this result is true for all positive integers 'n'. Hence proved.

Important Notes on Binomial Theorem

The following key points would help in a better understanding of the binomial theorem.

- The number of terms in the binomial expansion of $(x + y)^n$ is equal to $n + 1$.
- In the expansion of $(x + y)^n$, the sum of the powers of x and y in each term is equal to n .
- The value of the binomial coefficients from either side of the expansion is equal
- The number of terms in the binomial expansion of $(x + y + z)^n$ is $n(n + 1)$.

Example

Compute

$$(x + y)^4$$

Solution

Following the steps we outlined above, we find

$$\begin{aligned} (x + y)^4 &= \binom{4}{0} x^4 + \binom{4}{1} x^3 y + \binom{4}{2} x^2 y^2 + \binom{4}{3} x y^3 + \binom{4}{4} y^4 \\ &= x^4 + 4x^3 y + 6x^2 y^2 + 4x y^3 + y^4. \end{aligned}$$

Since $\binom{n}{0} = \binom{n}{n} = 1$, the expansion always starts with x^n and ends with y^n .

Example

$$(x - y)^4.$$

Solution

We find

$$\begin{aligned}(x - y)^4 &= [x + (-y)]^4 \\&= \binom{4}{0}x^4 + \binom{4}{1}x^3(-y) + \binom{4}{2}x^2(-y)^2 + \binom{4}{3}x(-y)^3 + \binom{4}{4}(-y)^4 \\&= x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + y^4.\end{aligned}$$

Take note of the alternating signs in the expansion. This suggests that we could expand

$$(A - B)^n$$

the exact same way we would with

$$(A + B)^n$$

except that the signs alternate.

We can carry out the expansion by following these steps. First, list all the terms we expect to find

$$(x + y)^4 = _ x^4 _ x^3y _ x^2y^2 _ xy^3 _ y^4.$$

Next, fill in the signs:

$$(x + y)^4 = _ x^4 - _ x^3y + _ x^2y^2 - _ xy^3 + _ y^4,$$

and then the binomial coefficients:

$$(x + y)^4 = \binom{4}{0}x^4 - \binom{4}{1}x^3y + \binom{4}{2}x^2y^2 - \binom{4}{3}xy^3 + \binom{4}{4}y^4.$$

Finally, compute the binomial coefficients to finish the expansion.

In-Text Questions (ITQs)

- Find the r th term in the expansion of

$$\left(x + \frac{1}{x}\right)^{2r}.$$

- Find the 4th term from the end in the expansion of

$$\left(\frac{x^3}{2} - \frac{2}{x^2}\right)^9$$

In-Text Answers (ITAs)

- We have

$$T_r = {}^{2r}C_{r-1} (x)^{2r-r+1} \left(\frac{1}{x}\right)^{r-1}$$

$$= \frac{{}^{2r}C_{r-1}}{x^{r+1-r+1}}$$

$$= \frac{{}^{2r}C_{r-1}}{x^2}$$

- Since r th term from the end in the expansion of $(a + b)^n$ is $(n - r + 2)$ th term from the beginning. Therefore 4th term from the end is $9 - 4 + 2$, i.e., 7th term from the beginning, which is given by

$$T_7 = {}^9C_6 \left(\frac{x^3}{2}\right)^3 \left(\frac{-2}{x^2}\right)^6 = {}^9C_3 \frac{x^9}{8} \cdot \frac{64}{x^{12}} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} \times \frac{64}{x^3} = \frac{672}{x^3}$$

10.1.2 Conclusion

Permutations and combinations provide powerful tools for counting arrangements and selections efficiently. Understanding the difference between ordered and unordered outcomes allows us to model and solve a wide range of problems in mathematics, computer science, and everyday decision-making. Mastery of factorials and selection formulas is foundational for topics in probability, logic, and algorithm design.

Summary

This unit covered the core principles of permutations and combinations, which are essential concepts in combinatorics and discrete mathematics. Permutations deal with ordered arrangements of objects, while combinations involve selections where the order does not matter. The factorial notation $n!$ was introduced to compute total permutations. Concepts such as bijective functions were connected to permutations, and the relationship between permutations and combinations was demonstrated mathematically. Examples highlighted the application of these formulas in various practical scenarios like committee selection and arranging books.

Self-Assessment Questions

1. What is the difference between a permutation and a combination?
2. How many 3-letter "words" can be formed from the letters {X, Y, Z, A, B} without repetition?

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. Find the value of $(x+y)^2$ and $(x+y)^3$ using Binomial expansion.
2. Find the 7th term in the expansion of $(x + 2)^{10}$

References/Further Readings

- Rosen, K. H. (2012). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill Education.
- Epp, S. S. (2010). Discrete Mathematics with Applications (4th ed.). Cengage Learning.
- Grimaldi, R. P. (2004). Discrete and Combinatorial Mathematics: An Applied Introduction (5th ed.). Pearson Education
- Kolman, B., & Busby, R. C. (2008). Discrete Mathematical Structures (6th ed.). Pearson.
- Tucker, A. (2002). Applied Combinatorics (4th ed.). John Wiley & Sons

Module Eleven

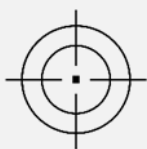
Discrete Probability

Unit 1 — Discrete Probability

INTRODUCTION

Closely related to the concepts of counting is Probability. We often try to guess the results of games of chance, like card games, slot machines, and lotteries; i.e. we try to find the likelihood or probability that a particular result will be obtained.

Probability can be conceptualized as finding the chance of occurrence of an event. Mathematically, it is the study of random processes and their outcomes. The laws of probability have a wide applicability in a variety of fields like genetics, weather forecasting, opinion polls, stock markets etc.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Define and explain core probability concepts including random experiments, sample space, events, and random variables.
- Distinguish between types of events (independent, dependent, mutually exclusive, equally likely) and random variables (discrete and continuous).
- Apply probability formulas, including conditional probability and Bayes' theorem, to solve basic and intermediate problems

11.1.1 Probability Theory

Probability theory is an advanced branch of mathematics that deals with measuring the likelihood of events occurring. It provides tools to analyze situations involving uncertainty and helps in determining how likely certain outcomes are. This theory uses the concepts of random variables, sample space, probability distributions, and more to determine the outcome of any situation.

For Example: Flipping a Coin

Flipping a coin is a random event with two possible outcomes: heads or tails. Each time you flip a fair coin, there are exactly two possible outcomes, each with an equal chance of occurring. Therefore, the probability of landing on heads is $1/2$, and similarly, the probability of landing on tails is also $1/2$.

11.1.2 Basics of Probability theory

Some important concepts of probability theory are:

Random Experiment

In probability theory, any event that can be repeated multiple times and whose outcome is not hampered by its repetition is called a Random Experiment.

For example, tossing a coin, rolling the dice, etc., are random experiments.

Sample Space

The set of all possible outcomes for any random experiment is called the sample space.

For example, throwing dice results in six outcomes, which are 1, 2, 3, 4, 5, and 6. Thus, its sample space is (1, 2, 3, 4, 5, 6)

Event

The outcome of any experiment is called an event. Various types of events used in probability theory are,

- **Independent Events:** The events whose outcomes are not affected by the outcomes of other future and/or past events are called independent events.
For example, the output of tossing a coin in repetition is not affected by its previous outcome.
- **Dependent Events:** The events whose outcomes are affected by the outcome of other events are called dependent events.

For example, picking oranges from a bag that contains 100 oranges without replacement.

- **Mutually Exclusive Events:** The events that cannot occur simultaneously are called mutually exclusive events.

For example, obtaining a head or a tail in tossing a coin, because both (head and a tail cannot be obtained together).

- **Equally likely Events:** The events that have an equal chance or probability of happening are known as equally likely events.

For example, observing any face in rolling the dice has an equal probability of $1/6$.

As the occurrence of any event varies between 0% and 100%, the probability varies between 0 and 1.

Random Variable

A variable that can assume the value of all possible outcomes of an experiment is called a random variable in Probability Theory. Random variables in probability theory are of two types, which are discussed below,

Discrete Random Variable

Variables that can take countable values, such as 0, 1, 2, ..., are called discrete random variables.

Examples: The number of heads when flipping 3 coins, the number of cars in a parking lot, or the number of correct answers on a test.

Continuous Random Variable

Variables that can take an infinite number of values in a given range are called continuous random variables.

Examples: The height of a person, the time it takes for a chemical reaction to occur, or the temperature of a substance.

11.1.3 Steps to finding Probability

Step 1 – Calculate all possible outcomes of the experiment.

Step 2 – Calculate the number of favourable outcomes of the experiment.

Step 3 – Apply the corresponding probability formula.

Tossing a Coin

If a coin is tossed, there are two possible outcomes – Heads (H) or Tails (T)

So, Total number of outcomes = 2

Hence, the probability of getting a Head (H) on top is $\frac{1}{2}$ and the probability of getting a Tails (T) on top is $\frac{1}{2}$

Throwing a Dice

When a dice is thrown, six possible outcomes can be on the top – 1,2,3,4,5,6

The probability of any one of the numbers is $\frac{1}{6}$

The probability of getting even numbers is $\frac{3}{6} = \frac{1}{2}$

The probability of getting odd numbers is $\frac{3}{6} = \frac{1}{2}$

Taking Cards From a Deck

From a deck of 52 cards, if one card is picked find the probability of an ace being drawn and also find the probability of a diamond being drawn.

Total number of possible outcomes – 52

Outcomes of being an ace – 4

Probability of being an ace = $\frac{4}{52} = \frac{1}{13}$

Probability of being a diamond = $\frac{13}{52} = \frac{1}{4}$

11.1.4 Probability Axioms

- The probability of an event always varies from 0 to 1. $[0 \leq P(x) \leq 1]$
- For an impossible event the probability is 0 and for a certain event the probability is 1.
- If the occurrence of one event is not influenced by another event, they are called mutually exclusive or disjoint.

If $A_1, A_2, \dots, A_n$ are mutually exclusive/disjoint events,

then $P(A_i \cap A_j) = 0$ for $i \neq j$ and $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$

Axiomatic probability is based on the formal, mathematical foundation established by Andrey Kolmogorov. It treats probability as a set function defined on a σ -algebra of events, satisfying a set of axioms (rules). This approach is highly abstract and provides a rigorous framework for probability theory.

Example: For a fair coin, there are two possible outcomes: Heads (H) or Tails (T). So, the sample space S is:

$$S = \{\text{Heads, Tails}\}$$

$$P(\text{Heads}) + P(\text{Tails}) = 1$$

So, using the axiomatic framework, we've established that: $P(\text{Heads}) = 0.5$, $P(\text{Tails}) = 0.5$

11.1.5 Properties of Probability

If there are two events x and x^c which are complementary, then the probability of the complementary event is –

$$p(\overline{x}) = 1 - p(x)$$

- For two non-disjoint events A and B , the probability of the union of two events –

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
- If an event A is a subset of another event B (i.e. $A \subset B$), then the probability of A is less than or equal to the probability of B . Hence, $A \subset B$ implies $P(A) \leq P(B)$

11.1.6 Conditional Probability

The conditional probability of an event B is the probability that the event will occur given an event A has already occurred. This is written as $P(B|A)$.

$$\text{Mathematically – } P(B|A) = \frac{P(A \cap B)}{P(A)}$$

If event A and B are mutually exclusive, then the conditional probability of event B after the event A will be the probability of event B that is $P(B)$

11.1.7 Probability Theory Formulas

Various formulas are used in probability theory, and some of them are discussed below,

- Theoretical Probability Formula: (Number of Favourable Outcomes) / (Number of Total Outcomes)

- Empirical Probability Formula: (Number of times event A happened) / (Total number of trials)
- Addition Rule of Probability: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Complementary Rule of Probability: $P(A') = 1 - P(A)$
- Independent Events: $P(A \cap B) = P(A) \cdot P(B)$
- Conditional Probability: $P(A | B) = P(A \cap B) / P(B)$
- Bayes' Theorem: $P(A | B) = P(B | A) \cdot P(A) / P(B)$ The conditional probability of an event B is the probability that the event will occur given an event A has already

11.1.8 Probability Theory in Statistics

Probability has various applications in Statistics. These are mentioned below:

- Descriptive Statistics: Probability theory helps in understanding and interpreting data summaries and distributions.
- Inferential Statistics: This forms the basis for making inferences about populations from samples, including hypothesis testing and the construction of confidence intervals.
- Regression Analysis: Probability distributions of errors are used to estimate the relationships between variables.
- Bayesian Statistics: Uses probability to represent uncertainty about the parameters of interest and updates this uncertainty as more data becomes available The conditional probability of an event B is the probability that the event will occur given an event A has already

In-Text Questions (ITQs)

1. In a country 50% of all teenagers own a cycle and 30% of all teenagers own a bike and cycle. What is the probability that a teenager owns bike given that the teenager owns a cycle?
2. A fair coin is tossed three times. What is the probability of getting exactly two heads?
3. Six good laptops and three defective laptops are mixed up. To find the defective laptops all of them are tested one-by-one at random. What is the probability to find both defective laptops in the first two pick?

In-Text Answers (ITAs)

Let us assume A is the event of teenagers owning only a cycle and B is the event of teenagers owning only a bike.

So, $P(A)=50/100=0.5$ and $P(A \cap B)=30/100$ from the given problem.

$$P(B|A)=P(A \cap B)/P(A)=0.3/0.5=0.6$$

Hence, the probability that a teenager owns bike given that the teenager owns a cycle is 60%.

Total possible outcomes when tossing a coin three times = $2^3 = 8$.
 Possible outcomes: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.
 Outcomes with exactly two heads: HHT, HTH, THH (3 outcomes).

Probability of getting exactly two heads:

$P(\text{exactly 2 heads}) = \text{Number of favorable outcomes} / \text{Total outcomes}$.

$$P(\text{exactly 2 heads}) = 3/8.$$

Let A be the event that we find a defective laptop in the first test and B be the event that we find a defective laptop in the second test.

$$\text{Hence, } P(A \cap B) = P(A)P(B|A) = 3/9 \times 2/8 = 1/12$$

11.1.9 Conclusion

Probability theory is a cornerstone of modern mathematics and statistics, offering tools to manage uncertainty and make informed predictions. Whether analyzing a simple coin toss or conducting complex statistical analyses, understanding probability equips students with critical problem-solving skills applicable across disciplines including science, engineering, finance, and data analysis.

Summary

This unit provided a thorough introduction to Probability Theory, a mathematical framework for quantifying uncertainty. Starting from basic concepts like sample spaces and events, the content progressed through classifications of random variables and types of events. It discussed methods for calculating probabilities using theoretical, empirical, and conditional approaches. Important rules and formulas,

such as the addition rule, complementary rule, and Bayes' Theorem. were introduced. Finally, it explained how probability theory underpins statistical practices such as hypothesis testing, regression, and Bayesian inference

Self -Assessment Questions

1. What is the difference between a discrete random variable and a continuous random variable?
2. Calculate the probability of drawing a red queen from a standard deck of 52 cards.
3. If the probability of event A is 0.7 and event B is 0.5, and A and B are independent, what is $P(A \cap B)$?

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)
3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1. Consider a jar with 7 red marbles, 3 green marbles, and 4 blue marbles. What is the probability of randomly selecting a non-blue marble from the jar?
2. Someone takes out one card from a 52-card deck, what is the probability of the card being a heart? What is the probability of obtaining a 7-number card?

References/Further Readings

Ross, S. M. (2014). Introduction to Probability Models (11th ed.). Academic Press.

Grinstead, C. M., & Snell, J. L. (1997). Introduction to Probability. American Mathematical Society.

Hogg, R. V., Tanis, E. A., & Zimmerman, D. L. (2018). Probability and Statistical Inference (10th ed.). Pearson.

DeGroot, M. H., & Schervish, M. J. (2011). Probability and Statistics (4th ed.). Pearson.

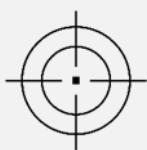
Module Twelve

Recurrence Relations

Unit 1 — Recurrence Relations

INTRODUCTION

Recurrence relations are a fundamental concept within the field of discrete mathematics, which itself is the study of mathematical structures that are fundamentally discrete rather than continuous. Unlike continuous mathematics, where quantities change smoothly, discrete mathematics deals with distinct and separate values. Within this context, recurrence relations serve as a pivotal tool to describe sequences and recursive structures.



LEARNING OUTCOMES

At the end of this unit, you should be able to:

- Define and explain core probability concepts including random experiments, sample space, events, and random variables.
- Distinguish between types of events (independent, dependent, mutually exclusive, equally likely) and random variables (discrete and continuous).
- Apply probability formulas, including conditional probability and Bayes' theorem, to solve basic and intermediate problems.

12.1.1 Recurrence Relation

A recurrence relation is an equation that recursively defines a sequence or multi-dimensional array of values. The relation specifies each term of the sequence as a function of the preceding terms. Understanding and solving such relations are crucial in the analysis of algorithms, computer science, combinatorics, and even economics.²

Formal Definition

A typical recurrence relation for a sequence a_n is of the form:

where f is a given function, and k is a fixed positive integer that determines the order of the recurrence.

Applications

Recurrence relations have numerous applications across various fields:

- Computer Science: In analyzing the time complexity of recursive algorithms.
- Combinatorics: Counting problems, such as the number of ways to arrange objects.
- Economics: Modeling economic processes and population growth.
- Engineering: Signal processing and systems theory.

Some Examples of Recurrence Relations

- a. The Fibonacci sequence is defined by the recurrence relation with the initial conditions $F_0=1$ and $F_1=1$. The recurrence relation is called a second-order relation because F_k depends on the two previous terms of F .
- b. The relation is a third-order recurrence relation. If values of $T(0)$, $T(1)$, and $T(2)$ are specified, then T is completely defined.
- c. The recurrence relation with $S(0)=0$ has infinite order. To determine $S(n)$ when n is even, you must go back $n/2$ terms. Since $n/2$ grows unbounded with n , no finite order can be given to S .

Solving Recurrence Relations

Sequences are often most easily defined with a recurrence relation; however, the calculation of terms by directly applying a recurrence relation can be time-consuming. The process of determining a closed form expression for the terms of a sequence from its recurrence relation is called solving the relation. There is no single technique or algorithm that can be used to solve all recurrence relations.

In fact, some recurrence relations cannot be solved. The relation that defines TT above is one such example. Most of the recurrence relations that you are likely to encounter in the future are classified as finite order linear recurrence relations with constant coefficients. This class is the one that we will spend most of our time within this chapter.

12.1.2 nth Order Linear Recurrence Relation

Let S be a sequence of numbers with domain $k \geq 0$. An n th order linear recurrence relation on S with constant coefficients is a recurrence relation that can be written in the form

$$S(k) + C_1 S(k-1) + \dots + C_n S(k-n) = f(k) \text{ for } k \geq n$$

where $C_1, C_2, \dots, C_n$ are constants and f is a numeric function that is defined for $k \geq n$.

Some Finite Order Linear Relations

- The Fibonacci sequence is defined by the second-order linear relation because

$$F_k - F_{k-1} - F_{k-2} = 0$$

- The relation

$$P(j) + 2P(j-3) = j^2$$

is a third-order linear relation. In this case, $C_1 = C_2 = 0$.

- The relation

$$A(k) = 2(A(k-1) + k)$$

can be written as

$$A(k) - 2A(k-1) = 2k.$$

Therefore, it is a first-order linear

Linear Recurrence Relations

A linear recurrence equation of degree k or order k is a recurrence equation which is in the format

$$x_n = A_1x_{n-1} + A_2x_{n-2} + A_3x_{n-3} + \dots + A_kx_{n-k} + A_n$$

is a constant and $A_k \neq 0$ on a sequence of numbers as a first-degree polynomial.

These are some examples of linear recurrence equations –

Recurrence relations	Initial values	Solutions
$F_n = F_{n-1} + F_{n-2}$	$a_1 = a_2 = 1$	Fibonacci number
$F_n = F_{n-1} + F_{n-2}$	$a_1 = 1, a_2 = 3$	Lucas Number
$F_n = F_{n-2} + F_{n-3}$	$a_1 = a_2 = a_3 = 1$	Padovan sequence
$F_n = 2F_{n-1} + F_{n-2}$	$a_1 = 0, a_2 = 1$	Pell number

Example

Solve the recurrence relation

$$F_n = 5F_{n-1} - 6F_{n-2} \text{ where } F_0 = 1 \text{ and } F_1 = 4$$

Solution

The characteristic equation of the recurrence relation is –

$$x^2 - 5x + 6 = 0,$$

$$\text{So, } (x-3)(x-2) = 0$$

Hence, the roots are –

$$x_1 = 3 \text{ and } x_2 = 2$$

The roots are real and distinct. So, this is in the form of case 1

Hence, the solution is –

$$F_n = ax_1^n + bx_2^n$$

Here,

$$F_n = a3^n + b2^n \text{ (As } x_1 = 3 \text{ and } x_2 = 2)$$

Therefore,

$$1 = F_0 = a3^0 + b2^0 = a + b$$

$$4 = F_1 = a3^1 + b2^1 = 3a + 2b$$

Solving these two equations, we get $a=2$ and $b=-1$

Hence, the final solution is –

$$F_n = 2.3^n + (-1).2^n = 2.3^n - 2^n$$

12.1.3 Algorithm for Solving Homogeneous Finite Order Linear Relations

- a) Write out the characteristic equation of the relation

$$S(k) + C_1 S(k-1) + \dots + C_n S(k-n) = 0, \text{ which is}$$

which is

$$a^n + C_1 a^{n-1} + \dots + C_{n-1} a + C_n = 0.$$

- b) Find all roots of the characteristic equation, the characteristic roots.

- c) If there are n distinct characteristic roots, $a_1, a_2, \dots, a_n$, then the general solution of the recurrence relation is

$$S(k) = b_1 a_1^k + b_2 a_2^k + \dots + b_n a_n^k.$$

If there are fewer than n characteristic roots, then at least one root is a multiple root. If a_j is a double root, then the $b_j a_j^k$ term is replaced with $(b_{j0} + b_{j1}k) a_j^k$. In general, if a_j is a root of multiplicity p , then the $b_j a_j^k$ term is replaced with

$$S(k) = b_1 a_1^k + b_2 a_2^k + \dots + b_n a_n^k.$$

- d. If n initial conditions are given, we get n linear equations in n unknowns (the b_j 's from Step 3) by substitution. If possible, solve these equations to determine a final form for $S(k)$.

Although this algorithm is valid for all values of n , there are limits to the size of n for which the algorithm is feasible. Using just a pencil and paper, we can always solve second-order equations. The quadratic formula for the roots of $ax^2 + bx + c = 0$ is

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The solutions of

$$a^2 + C_1 a + C_2 = 0$$

Are then

$$\frac{1}{2}(-C_1 + \sqrt{C_1^2 - 4C_2}) \text{ and } \frac{1}{2}(-C_1 - \sqrt{C_1^2 - 4C_2})$$

Although cubic and quartic formulas exist, they are too lengthy to introduce here. For this reason, the only higher-order relations ($n \geq 3$) that you could be expected to solve by hand are ones for which there is an easy factorization of the characteristic polynomial.

12.1.4 Solution of Nonhomogeneous Order Linear Relations

A recurrence relation is called non-homogeneous if it is in the form

$$F_n = AF_{n-1} + BF_{n-2} + f(n) \text{ where } f(n) \neq 0$$

Its associated homogeneous recurrence relation is $F_n = AF_{n-1} + BF_{n-2}$

The solution (an) of a non-homogeneous recurrence relation has two parts.

First part is the solution (ah) of the associated homogeneous recurrence relation and the second part is the particular solution (at).

$$a_n = a_h + a_t$$

Solution to the first part is done using the procedures discussed in the previous section.

To find the particular solution, we find an appropriate trial solution.

Let

$$f(n) = cx^n ; \text{ let } x^2 = Ax + B$$

be the characteristic equation of the associated homogeneous recurrence relation and let x_1 and x_2 be its roots.

- If $x \neq x_1$ and $x \neq x_2$, then $a_t = Ax^n$
- If $x = x_1, x \neq x_2$, then $a_t = Anx^n$
- If $x = x_1 = x_2$, then $a_t = An^2x^n$

Example

Let a non-homogeneous recurrence relation be

$$F_n = AF_{n-1} + BF_{n-2} + f(n)$$

with characteristic roots $x_1=2$ and $x_2=5$. Trial solutions for different possible values of $f(n)$ are as follows –

$f(n)$	Trial solutions
4	A
5.2^n	$An2^n$
8.5^n	$An5^n$
4^n	$A4^n$

In-Text Questions (ITQs)

1. Solve the recurrence relation

$$F_n = 10F_{n-1} - 25F_{n-2} \text{ where } F_0 = 3 \text{ and } F_1 = 17$$

2. Solve the recurrence relation

$$F_n = 3F_{n-1} + 10F_{n-2} + 7.5^n \text{ where } F_0 = 4 \text{ and } F_1 = 3$$

In-Text Answers (ITAs)

1. The characteristic equation of the recurrence relation is –

$$x^2 - 10x - 25 = 0$$

$$\text{So } (x - 5)^2 = 0$$

Hence, there is single real root $x_1=5$

As there is single real valued root, this is in the form of case 2

Hence, the solution is –

$$F_n = ax_1^n + bnx_1^n$$

$$3 = F_0 = a.5^0 + (b)(0.5)^0 = a$$

$$17 = F_1 = a.5^1 + b.1.5^1 = 5a + 5b$$

Solving these two equations, we get $a=3$ and $b=2/5$

Hence, the final solution is

2. This is a linear non-homogeneous relation, where the associated homogeneous equation is

The characteristic equation of its associated homogeneous relation is –

$$x^2 - 3x - 10 = 0$$

$$\text{Or, } (x - 5)(x + 2) = 0$$

$$\text{Or, } x_1 = 5 \text{ and } x_2 = -2$$

Hence $a_h = a.5^n + b.(-2)^n$, where a and b are constants.

12.1.5 Conclusion

Understanding recurrence relations is essential in analyzing recursive algorithms and modeling real-world problems in computer science and engineering. With mastery of characteristic equations and solution algorithms, learners can confidently solve a wide range of recurrence-based problems, enhancing their analytical and computational skills.

Summary

This unit covered recurrence relations, which are equations that define sequences based on preceding terms. Emphasis was placed on linear recurrence relations with constant coefficients, including how to solve them using characteristic equations. The solution methods for both homogeneous and non-homogeneous cases were presented, including step-by-step examples. Applications in computer science, engineering, and combinatorics were also discussed.

Self -Assessment Questions

1. What distinguishes a homogeneous recurrence relation from a non-homogeneous one?
2. Find a recurrence relation and initial conditions for 1,5,17,53,161,485...

Unit Activities

1. Read the content of Unit 1 in the study guide. Consult the recommended texts and research more information on the internet. (90 minutes)
2. Complete the self-introductory forum. Original post by Thursday. (10 minutes). Complete replies to 2 of your classmates by Sunday at 11:59 p.m. (10 minutes)

3. TUTOR MARKED ASSIGNMENT: Complete the following and submit for grading. (60 minutes)

Tutor-Marked Assignment

1.
Check that $a_n = 2^n + 1$ is a solution to the recurrence relation $a_n = 2a_{n-1} - 1$ with $a_1 = 3$.
2.
Solve the recurrence relation $a_n = a_{n-1} + n$ with initial term $a_0 = 4$.

References/Further Readings

- Rosen, K. H. (2019). Discrete Mathematics and Its Applications (8th ed.). McGraw-Hill Education.
- Grimaldi, R. P. (2003). Discrete and Combinatorial Mathematics: An Applied Introduction (5th ed.). Pearson Education.
- Epp, S. S. (2011). Discrete Mathematics with Applications (4th ed.). Cengage Learning.
- Levitin, A. (2011). Introduction to the Design and Analysis of Algorithms (3rd ed.). Pearson Education.

Answers to Self-Assessment Questions

GLOSSARY

